

External Negotiations and Customs Union Stability

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September 28, 2026

Abstract

Mercosur has faced exit threats for years, but no withdrawal has occurred. This paper studies that pattern in Mercosur, the largest customs union in South America, and asks why membership can remain valuable while new external agreements become difficult to approve. I show that Mercosur's customs-union structure can make external bargaining more valuable for a small member while making new agreements harder to approve inside the bloc. To do so, I separate the value of bargaining as a bloc from the difficulty of getting every member to support a new external tariff agreement. I build a structural model with reciprocal external bargaining and asymmetric applied tariffs, calibrated to sectoral trade data for 5 different years between 2004 and 2017. In the focal China and United States exercises, Uruguay gains more from bloc bargaining than from bargaining alone, while Brazil and Argentina obtain smaller or negative aggregate effects and face concentrated sectoral losses. The results provide a mechanism consistent with Mercosur remaining valuable enough for members to stay while still creating approval fragility around external liberalization.

JEL Codes: F13, F15, F14, F02, D78.

Keywords: Customs Unions, Trade Agreements, Reciprocity, Mercosur, External Negotiations, Quantitative Trade.

1 Introduction

Mercosur's recent history illustrates the central problem studied in this paper: a customs union can preserve enough collective value to keep its members inside while the same collective decision rules make external liberalization difficult to approve. When negotiations with large partners move slowly, members that want more access to external markets may ask whether they could gain negotiating flexibility outside the bloc. Uruguay's discussions with the United States in 2004–2007, its later China initiative, and the EU–Mercosur negotiation process all point to this tension. Yet no founding member has withdrawn. The question is therefore not only why external agreements are delayed, but why the bloc can remain stable even when some members see more flexible external negotiations as attractive.

To understand this problem, the paper separates the member's decision to remain in the customs union from the government's decision to approve a particular external agreement. On the participation margin, a member evaluates bargaining through Mercosur relative to a specified outside option; on the approval margin, each government evaluates a proposed external tariff change relative to the existing common external policy. Separating these comparisons organizes the analysis without assuming that the two decisions are independent in practice. It allows the model to capture a case in which Uruguay values the bargaining scale of the bloc, while Brazil and Argentina may have weaker incentives to approve reciprocal tariff cuts that expose import-competing manufacturing sectors.

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I am grateful for all the guidance and support of Mostafa Beshkar, Gustavo Torrens, Volodymyr Lugovskyy, and Sarah Bauerle-Danzman. I also thank participants at the Midwest Trade and Theory Conference and the Indiana University Microeconomics Seminar for valuable feedback. I thank the Indiana University Economics Department for funding support provided for this research. I would like to also thank Tulio Sousa, Marco Acosta Carrion, Rodrigo Ito, Mauricio S. Bugarin, and Junho Kim.

Before attributing bargaining value to Mercosur's common external policy, I examine the protection that remains under national exceptions and the way members negotiate with outside partners. The country reports distinguish departures from the CET from goods that already face a common zero rate. The agreement documents show how Mercosur exchanges access to its market for access abroad. Together, this evidence motivates comparing bloc and single-member negotiations under both reciprocal tariff cuts and full bilateral tariff elimination.

The quantitative exercise then asks how those margins look when the same tariff environment is evaluated under different bargaining modes. I embed reciprocal bargaining in an Ossa-style Krugman–Armington (Ossa (2016)) environment calibrated to sectoral trade and tariff data for 2004, 2007, 2011, 2014, and 2017 with GTAP 11 data. The quantitative results use the WITS-main calibration. The reciprocity criterion selects tariff cuts that minimize the imbalance under a specified measure of market-access concessions. I define concessions as the fall in real net exports from the applied-tariff baseline and relate this operational measure to the balance-of-concessions literature in Section 3.1.1. I compare bloc bargaining, single-member bargaining, and outside-option benchmarks directly under reciprocal tariff cuts and zero-tariff benchmarks (full liberalization of tariffs), while retaining exclusion-style outside options as auxiliary comparisons.

Once the model maps tariff changes into welfare and sectoral exposure, the counterfactuals point to different incentives across members. In the China 2017 reciprocal exercise, Uruguay gains 0.598 percent as part of Mercosur and loses 0.232 percent when bargaining alone under the same rule, yielding a bloc-minus-own-member difference of 0.830 percentage points. In the United States 2004 reciprocal exercise, Uruguay gains 0.867 percent as part of Mercosur and loses 0.006 percent alone, yielding a difference of 0.874 percentage points. I interpret these differences as bargaining-value components of the participation margin, not as complete estimates of the legal value of membership. For Brazil and Argentina, the same exercises produce smaller or negative aggregate effects. In China 2017, reciprocal bloc gains are 0.131 and 0.094 percent; in United States 2004, they are -0.111 and -0.077 percent. The paper therefore finds that participation and approval incentives can differ: a customs union may remain valuable for membership while a proposed external agreement remains fragile when aggregate effects are small and losses are concentrated.

These comparisons extend Mercosur counterfactual studies that evaluate the welfare consequences of particular agreements or partner choices (Caliendo and Parro, 2010; Timini and Viani, 2020). I compare the same member, partner and benchmark economy under collective and single-member negotiations, and distinguish the value of bargaining through the bloc from a member's incentive to approve a particular external agreement. Evaluating reciprocal cuts alongside full bilateral tariff elimination allows these comparisons to depend on the concessions exchanged and the protection that remains. The institutional evidence establishes the setting for these exercises, while the model measures their welfare and sectoral consequences.

Related Literature: The argument connects four strands of trade research because each one speaks to a different part of the participation and approval problem. Customs-union theory explains why joint external policy and collective bargaining can create value for members. Political-economy and self-enforcement work clarifies why agreements that look attractive in aggregate may still be difficult to sustain or approve. Quantitative trade and reciprocity provide the tools for comparing bargaining modes under observed trade and tariff data. Mercosur-specific empirical work supplies the institutional setting in which these margins can be studied.

Classic customs-union theory begins with the welfare ambiguity created by trade creation and trade diversion (Viner, 1982). Meade (1955) and Lipsey (1957) formalize the general-equilibrium trade-off, while Kemp and Wan (1976) shows that a suitable common external tariff and transfers can make members better off without harming outsiders. For Mercosur, the transfer logic is important precisely because compensation is limited in practice: aggregate gains do not automatically imply approval. Work on coalition stability and regionalism then links these welfare effects to participation incentives (Riezman, 1985; Krishna, 1998), and Bagwell and Staiger (1997) connects self-enforcement to terms-of-trade externalities and reciprocal agreements.

More recent customs-union research shifts attention from static welfare rankings to political and dynamic constraints. [Facchini et al. \(2013\)](#) show how coordinating external tariffs can distort political delegation (and be distorted by it), and [Facchini et al. \(2021\)](#) show that trade imbalances, inequality, and production differences affect whether countries form preferential agreements and which institutional form they choose. This point is directly relevant here because Mercosur's common external tariff is not only a tariff schedule; it is a policy instrument over which heterogeneous members and sectors bargain.

The common external tariff also connects external bargaining to internal distributional conflict. [Krueger \(1997\)](#) clarifies the institutional difference between free trade areas and customs unions. [Ornelas \(2007\)](#) shows how customs-union members can coordinate external tariffs to exchange market access and shift rents from outsiders. [Syropoulos \(2003\)](#) studies how tariff-revenue rules affect members' preferred common external tariff, and [Olarreaga et al. \(1999\)](#) provides Mercosur evidence that both terms-of-trade forces and interest-group pressure shaped the bloc's tariff structure. Recent Latin American evidence further shows that preferential margins differ systematically between customs unions and free trade areas ([Ornelas and Tovar, 2022](#)). I use this literature to treat common external policy as the link between bargaining power and approval conflict.

A related dynamic literature emphasizes the flexibility cost of customs unions. [Lake \(2019\)](#) frames the choice between a customs union and a free trade area as a trade-off between coordinated external policy and the ability to form future agreements independently. [Lake and Yildiz \(2016\)](#) helps explain why customs unions tend to be regional, while [Lake \(2017\)](#) separates building-block and stumbling-block mechanisms. I build on these insights but study a different setting: an existing customs union that can remain stable because of bloc bargaining value while becoming gridlocked because approval margins differ across members.

Because the paper interprets approval through government preferences and sectoral pressure, it also draws on the traditional "protection-for-sale" literature of trade. [Grossman and Helpman \(1994\)](#) provide the canonical protection-for-sale framework, and [Maggi and Rodríguez-Clare \(1998\)](#) shows why trade agreements matter when governments face domestic political pressure. The mapping from sectoral losses to policy pressure need not be linear because organization costs and firm-size concentration differ across sectors ([Mitra, 1999](#); [Bombardini, 2008](#)). I use this literature to interpret concentrated sectoral losses as evidence of weak approval margins, not as estimated lobbying contributions. The paper connects tariff bargaining to a simple political-economy point: uneven member-level gains can make consensus approval harder even when the bloc gains in aggregate.

Finally, the paper contributes to quantitative trade work on policy counterfactuals. In quantitative trade theory, [Caliendo and Parro \(2015\)](#) and [Ossa \(2016\)](#) provide tractable frameworks for evaluating tariff changes using observed trade shares and elasticities. [Bagwell et al. \(2021\)](#) and [Beshkar et al. \(2025\)](#) bring reciprocity and balanced concessions into quantitative trade analysis, building on the exact-hat-algebra logic of [Dekle et al. \(2007\)](#). Related work on import market power and trade agreements, including [Beshkar and Lee \(2022\)](#), motivates the focus on tariff bargaining and market-access concessions. Within the Mercosur literature, [Caliendo and Parro \(2010\)](#) evaluates Uruguay's partner options, [Campos and Timini \(2022\)](#) studies unequal gains across members, and [Timini and Viani \(2020\)](#) analyzes EU–Mercosur effects. I differ by comparing bargaining modes directly and by using those comparisons to separate internal stability from external approval gridlock.

2 Institutional Framework

The institutional evidence addresses two questions: how national exceptions affect the protection associated with the CET, and why reciprocal exchanges of concessions are a realistic basis for Mercosur negotiations. I use the country reports to answer the first question and the bloc's legal and agreement documents to answer the second.

2.1 Coverage and implementation of the common external tariff

The relevance of the CET depends on the protection it maintains and the rules governing departures from it. A zero national rate can simply implement a zero CET rate, or it can be an exception to a positive CET rate. The second case can address a production constraint: Brazil's product-specific concessions for capital and information-technology goods allow zero rates when no equivalent domestic production is identified ([Ministry of Development, Industry, Trade and Services of Brazil, 2024](#)). A national rate below the CET can also remain positive. These cases have different implications for protection, so a single percentage of goods or imports "covered by the CET" cannot establish how much economically relevant policy the bloc coordinates.

I assess national flexibility by reading the official product lists together with the bloc rules and the measures that implement them. Decision 58/10 authorizes national lists, limits changes to their composition, and requires consideration of Mercosur's export supply ([MERCOSUR Common Market Council, 2010](#)). Uruguay's Decree 237/2017 uses that framework to add, remove and amend exceptions ([Government of Uruguay, 2017](#)). National differences therefore do not by themselves establish independent tariff setting: some operate through jointly agreed permissions. Appendix A.6.3 explains how I retain each list's date, rates and eligibility conditions, and distinguish a newly granted reduction from a change in its scope.

The country reports document national departures that extend beyond a few isolated products. The recovered national exception lists contain 92 below-CET entries in Argentina, 320 in Brazil and 690 in Paraguay. Capital goods, information technology and other arrangements add further departures. In Uruguay's February 2025 national schedule, 4,204 of 12,243 published entries are below the CET, or 34.3 percent. Table 16 in Appendix A.6.3 gives the source dates, regimes and counting units for all four countries. These are counts of listed tariff treatment. Because entries can repeat a product code and the national collections differ, they cannot be added into a bloc total or used to rank the share of imports receiving exemptions.

Uruguay's detailed classification shows why departures from the CET cannot be read as an absence of protection. Of its 4,204 below-CET entries, 3,205 retain a positive national tariff. Another 4,961 entries apply a positive CET rate, while 3,008 are already zero under the common schedule (Table 17). The economic importance of these rates depends on the goods traded and the production they protect, including production in other Mercosur members. The inventories document this remaining protection and the scope of national discretion; they do not measure the share of regional production protected or the effect of coordination on trade. The model therefore starts from observed applied tariffs, allowing national differences, and evaluates the consequences of further tariff concessions. Regional production also affects the interpretation of tariff relief. A good not produced in the importing member may be supplied by another Mercosur member. Reducing its external tariff can narrow that supplier's preference even when qualifying intrabloc imports already enter duty free. Rules of origin determine access to those preferences, while automotive and sugar trade retain special arrangements ([MERCOSUR Common Market Council, 2023](#)). The supplier comparisons in Appendix A.6.3 therefore distinguish changes in regional sales from a new tariff cut for regional goods.

2.2 External negotiations and tariff liberalization

Mercosur's agreement with India and its negotiating framework with Japan show how an exchange of concessions enters the bloc's trade policy. The 2004 Mercosur–India agreement explicitly describes reciprocal tariff preferences in its preamble. Article 3 assigns the preferences offered by Mercosur to Annex I and those offered by India to Annex II ([Mercosur and India, 2004](#), Preamble and Article 3(a)–(b)). The June 2026 Mercosur–Japan framework sets out a negotiating objective: paragraph 5 calls for a balanced, mutually beneficial agreement that considers both sides' sensitivities, with all goods subject to negotiation ([Mercosur, 2026b](#), paragraph 5). The India agreement thus provides for concessions by both parties, while the Japan framework states the balance sought in a future agreement.

This approach is consistent with the reciprocity principle expressed in GATT Article XXVIII bis(1), which describes tariff negotiations on a “reciprocal and mutually advantageous basis.” Paragraph 3 also requires negotiations to allow consideration of countries’ and industries’ needs, including development needs ([World Trade Organization, 1994](#), Article XXVIII bis(1), (3)). Neither cited Mercosur passage invokes this provision. I interpret their exchange of preferences and balanced-agreement language as evidence that reciprocity guides the negotiations, rather than as proof that GATT requires a Mercosur–China agreement to balance concessions. The provision supplies no mandatory formula for equal tariff cuts, equal welfare gains or the numerical balance used here.

The need to obtain agreement within Mercosur makes the content of that exchange important for each member. The Treaty of Asuncion establishes common external commercial policy, while the Protocol of Ouro Preto assigns negotiating authority and requires consensus in Mercosur decisions ([MERCOSUR, 1991, 1994](#)). A common offer must therefore accommodate members with different export opportunities and import-competing sectors. This is the institutional basis for asking both whether bargaining through the bloc is valuable and whether each member benefits from a particular exchange of concessions.¹

Uruguay’s China initiative makes the choice between a national and a bloc offer relevant to the application. The two governments confirmed completion of a feasibility study in 2022, and their February 2026 joint declaration expressed an aspiration to begin Mercosur–China trade negotiations ([Presidency of Uruguay, 2022](#); [Ministry of Commerce of China, 2022](#); [Governments of Uruguay and China, 2026](#)). The reviewed documents contain no agreed tariff schedule. I therefore compare hypothetical exchanges under the two negotiating parties; Appendix A.6 preserves the diplomatic chronology and distinguishes trade negotiations from bilateral investment work.

I represent this exchange through two tariff exercises. Full bilateral liberalization sets both sides’ affected tariffs to zero. The reciprocal-cut exercise instead selects cuts that minimize the imbalance under the model’s measure of concessions, allowing the sides to make different percentage cuts. Full liberalization is also a two-sided agreement; the distinction is whether all remaining tariffs are removed or their reductions are selected using the concession measure. The formula and restricted search in Section 3.1.1 are modeling choices for studying that exchange. Comparing both exercises under national and bloc offers then shows how negotiating scale and the chosen concessions affect member outcomes.

3 Theoretical Framework

The model formalizes the institutional distinction at the core of this paper: a member may prefer to remain inside a customs union and still prefer the status quo to a proposed external agreement. In that case, the union can remain internally stable while becoming externally stagnant from the perspective of a small member seeking more flexible liberalization. The participation margin compares bargaining through the customs union with a specified outside option. The approval margin asks whether each member gains from the proposed external tariff change relative to the status quo. Separating these margins allows the same customs union to be valuable for participation while still facing gridlock over external liberalization.

3.1 Economic environment

To quantify these margins, I use an Ossa-style Krugman–Armington CES model with markups ([Ossa, 2016](#)). The model maps tariff changes into prices, trade flows, profits, and welfare through terms-of-trade and profit-shifting effects that can differ across countries and sectors. It is solved in exact hat algebra, so the counterfactuals can be computed from observed trade shares, expenditure shares, tariffs, and sectoral elasticities. The appendix reports the core data, aggregation, selected reciprocal tariff cuts, and calibration details used in the counterfactuals.

¹Decision 32/00 reaffirms joint negotiation of tariff preferences. Its official database record lists an incorporation requirement, no incorporation by the four founding members and no recorded entry-into-force date ([MERCOSUR Common Market Council, 2000](#); [Mercosur, 2026a](#)). The authority and consensus argument here also rests on the founding treaties.

The sectoral structure is needed to interpret the approval margin because tariff changes affect governments through industry-level import competition, export access, and profit-shifting. Firms in country i and sector s produce with unit labor requirement $\varphi_{i,s}$ and charge a constant markup $\sigma_s/(\sigma_s - 1)$ over marginal cost. Goods move across borders subject to iceberg costs and ad valorem tariffs, so importer j faces delivered prices that depend on both iceberg costs and the gross tariff factor $(1 + t_{ij,s})$. This structure implies a CES sectoral price index and expenditure switching across origins.

Geographically, the world consists of N countries indexed by $i, j \in \{1, \dots, N\}$ and S sectors indexed by $s \in \{1, \dots, S\}$. Throughout, the first country index is the exporter and the second is the importer: $t_{ij,s}$ is the ad valorem tariff applied by country j to imports from country i in sector s . In each country j , a representative consumer allocates Cobb–Douglas expenditure shares $\mu_{j,s}$ across sectors, and within each sector aggregates country varieties with CES elasticity $\sigma_s > 1$. Let $P_{j,s}$ denote the sectoral price index and $P_j = \prod_s P_{j,s}^{\mu_{j,s}}$ the aggregate price index.

The iceberg costs $\tau_{ij,s}$ and ad valorem tariffs $t_{ij,s}$ jointly form the delivered trade cost $\kappa_{ij,s} = (1 + t_{ij,s})\tau_{ij,s}$. Hence, within each sector, CES demand implies standard gravity-style expenditure shares. Let $p_{ij,s}$ denote the delivered price, including tariffs, in j for a variety from i . Write $x_{ij,s}$ for producer-valued trade and $X_{ij,s} = (1 + t_{ij,s})x_{ij,s}$ for tariff-inclusive expenditure. Sectoral expenditure shares therefore satisfy

$$\lambda_{ij,s} \equiv \frac{X_{ij,s}}{\sum_k X_{kj,s}} = \frac{(p_{ij,s})^{1-\sigma_s}}{\sum_k (p_{kj,s})^{1-\sigma_s}}. \quad (1)$$

On the supply side, firms produce differentiated varieties under monopolistic competition and charge a constant markup $\sigma_s/(\sigma_s - 1)$ over marginal cost. In reduced form, delivered prices can be written as

$$p_{ij,s} = \frac{\sigma_s}{\sigma_s - 1} (1 + t_{ij,s}) \tau_{ij,s} c_{i,s}, \quad (2)$$

where $c_{i,s}$ is the marginal cost index (proportional to the wage under a one-factor technology). Holding the rest of the environment fixed, a tariff change $t_{ij,s}$ affects welfare through (i) the standard price/expenditure-switching channel (via the sectoral price index $P_{j,s}$ and therefore real income), and (ii) a profit channel, because markups imply that changes in market access and competitive pressure shift operating profits across countries and sectors. For the participation exercises, a useful way to summarize the additional mechanism relative to a perfectly competitive Armington model is via a decomposition into a terms-of-trade component and a profit-shifting component, following [Ossa \(2016\)](#). Under perfect competition, profits are zero by construction, so the profit-shifting channel is shut down; under monopolistic competition, it is active and can generate concentrated industry-level losses even when aggregate real income increases. The exercises below keep this market structure fixed: they do not model endogenous firm entry, zero-profit free entry, or later-period strategic responses after an exit or agreement shock.

The comparison evaluates counterfactual tariff vectors using exact hat algebra. Given sectoral elasticities and baseline expenditure/trade shares, the counterfactual equilibrium delivers changes in wages, sectoral price indices, and bilateral trade flows, from which we compute real-income (welfare) changes. The appendix reports the data, selected reciprocal tariff cuts, and calibration details used in these fixed-environment tariff counterfactuals.

3.1.1 Selection of reciprocal tariff cuts

Reciprocal exercises select tariff cuts by minimizing the imbalance in concessions between the two bargaining sides. Let $\mathcal{B} \subseteq \mathcal{M}$ denote the Mercosur-side bargaining set and let e denote the external partner. In a bloc negotiation, $\mathcal{B} = \mathcal{M}$. In a single-member negotiation, $\mathcal{B} = \{m\}$ for member m . Let $\rho_{\mathcal{B}}, \rho_e \in [0, 1]$ denote the proportional tariff cuts chosen by the Mercosur side and by the external partner.

I use this criterion to select a hypothetical exchange of tariff concessions and then evaluate its consequences for the negotiating members. A small imbalance under the chosen measure can coexist with different welfare changes across members, so selection precedes the assessment of participation and approval. Services, investment, regulatory commitments and implementation schedules may also affect the acceptability of an actual agreement. Isolating the tariff exchange allows the economic consequences of collective and single-member bargaining to be compared under the same rule.

For affected bilateral tariffs, candidate cuts map the same-year baseline tariff vector $\mathbf{t}^0$ into a counterfactual tariff vector $\tilde{\mathbf{t}}(\rho_{\mathcal{B}}, \rho_e)$:

$$\tilde{t}_{ej,s}(\rho_{\mathcal{B}}) = (1 - \rho_{\mathcal{B}})t_{ej,s}^0, \quad j \in \mathcal{B}, \quad \tilde{t}_{ie,s}(\rho_e) = (1 - \rho_e)t_{ie,s}^0, \quad i \in \mathcal{B}. \quad (3)$$

All other tariffs remain at their same-year applied baseline values.

Let $NX_k(\mathbf{t})$ denote worldwide real net exports of bargaining side k under tariff vector $\mathbf{t}$, for $k \in \{\mathcal{B}, e\}$. I exclude domestic flows and divide each producer-valued trade flow by the importing country's sectoral price hat before summing exports minus imports. This is a within-benchmark normalization, not a common price-year deflator. For the Mercosur bargaining side, define

$$NX_{\mathcal{B}}(\mathbf{t}) = \sum_{m \in \mathcal{B}} NX_m(\mathbf{t}). \quad (4)$$

The operational concession measure is the fall in real net exports relative to the baseline,

$$C_k(\rho_{\mathcal{B}}, \rho_e) = - \left[NX_k(\tilde{\mathbf{t}}(\rho_{\mathcal{B}}, \rho_e)) - NX_k(\mathbf{t}^0) \right], \quad k \in \{\mathcal{B}, e\}. \quad (5)$$

This measure provides an operational score for comparing tariff exchanges from the observed policy baseline. It is motivated by the balance-of-concessions literature ([Bagwell and Staiger, 1999](#); [Beshkar et al., 2025](#)).² It is not a legal valuation of concessions. The concession imbalance is

$$\mathcal{R}_{\mathcal{B},e}(\rho_{\mathcal{B}}, \rho_e) = C_{\mathcal{B}}(\rho_{\mathcal{B}}, \rho_e) - C_e(\rho_{\mathcal{B}}, \rho_e). \quad (6)$$

A positive value means that the Mercosur-side bargaining set gives the larger concession under this measure. The selected reciprocal tariff cuts are

$$(\rho_{\mathcal{B}}^*, \rho_e^*) \in \arg \min_{(\rho_{\mathcal{B}}, \rho_e) \in \mathcal{S}_{\mathcal{B},e}} |\mathcal{R}_{\mathcal{B},e}(\rho_{\mathcal{B}}, \rho_e)|, \quad (7)$$

where $\mathcal{S}_{\mathcal{B},e}$ contains 199 positive boundary pairs on a one-percentage-point grid, with at least one side cutting its affected tariffs by 100 percent. Exact ties favor the larger sum of cuts, then the larger Mercosur/member cut, then the larger partner cut. The search includes neither interior pairs nor a no-agreement option and imposes no welfare-improvement filter. The schedules were selected under each benchmark's original `SIGMA.csv` vector and are held fixed when welfare is evaluated under WITS and the alternative calibrations; these are not newly optimized WITS bargains. The stored scores for all 50 negotiations reproduce the selected schedules when re-ranked, although the 9,950 score-generating equilibria have not been freshly recomputed. The zero-tariff benchmark is the special counterfactual that sets the affected bilateral tariffs to zero:

$$\tilde{t}_{ej,s} = 0, \quad j \in \mathcal{B}, \quad \tilde{t}_{ie,s} = 0, \quad i \in \mathcal{B}.$$

²I cite the December 11, 2025 version of [Beshkar et al. \(2025\)](#), which relates concessions to multilateral terms-of-trade gains and net-import changes at fixed prices. Here flows are normalized by importer-sector price changes and compared with the applied-tariff baseline. I do not assume that this normalization and restricted tariff search reproduce that paper's fixed-price measure or its noncooperative benchmark.

3.2 Customs union stability and external negotiation gridlock

My argument for this paper starts from the common external tariff as a bargaining instrument. Mercosur's collective negotiating framework limits independent preferential commitments while allowing members to offer access to a larger market. National exceptions make applied protection uneven, so a common negotiating framework need not imply identical tariffs across members (Section 2). For a small member such as Uruguay, this can raise the value of participation because access to the Mercosur market is bundled into the external negotiation. At the same time, the same common external tariff makes liberalization a collective decision over sectors that are protected differently across members. A proposed agreement can therefore increase Uruguay's bargaining value while giving Brazil or Argentina weaker approval incentives if the tariff cuts reduce protection in politically salient manufacturing sectors. This is the specific trade-off modeled below: joint external policy can support internal stability through bargaining scale, while also creating an approval constraint when the distributional effects of external liberalization differ across members. This logic builds on the distinction between free trade agreements and customs unions in [Krueger \(1997\)](#), the bargaining role of the common external tariff in [Ornelas \(2007\)](#), and the importance of tariff-revenue rules and member heterogeneity in [Syropoulos \(2003\)](#). For Mercosur, [Olarreaga et al. \(1999\)](#) show that the common external tariff reflected both economic incentives and political-economy pressures, while [Ornelas and Tovar \(2022\)](#) show that preferential margins differ systematically across agreement types. For each scenario, I evaluate the counterfactual trade equilibrium implied by the policy alternative and use $G_i(\cdot)$ to denote the policy value of member i , including real-income effects and political effects from sectoral gains and losses.

3.2.1 Bloc bargaining power and internal stability

Joining a customs union changes a member's outside options in two directions at once. Mercosur's collective external-policy framework limits the scope for independent preferential tariff commitments, while bargaining as a bloc pools members' offers of market access. This flexibility–bargaining-power duality is central to the dynamic PTA logic in [Lake \(2019\)](#), where institutional flexibility affects the formation and value of future trade agreements. [Lake and Yildiz \(2016\)](#) makes the related point that free trade agreements and customs unions organize future bargaining differently and therefore have different institutional and geographic patterns. I add to this debate by taking that institutional trade-off inside an existing customs union and separating two margins that are often bundled together. The participation margin asks whether a member, such as Uruguay, values customs-union bargaining more than a specified outside option, even though membership restricts unilateral flexibility. The approval margin asks whether every member government prefers a proposed external agreement to the status quo, which can fail when reciprocal tariff cuts expose protected sectors in Brazil or Argentina. The contribution is therefore to show how the same institutional feature that can support internal stability through bargaining scale can also generate external gridlock through member-level approval constraints.

The zero-tariff counterfactuals separate the extent of liberalization from the choice of negotiating party. Full bilateral elimination removes the affected tariffs on both sides, whether the agreement is negotiated by Mercosur or by a single member. The reciprocal-cut exercise selects reductions using the concession measure in Section 3.1.1. Both exercises exchange market access; only the latter uses that measure to select the cuts. Comparing bloc and single-member outcomes within each exercise then isolates the value of negotiating through Mercosur under the specified tariff arrangement. The participation comparison therefore uses a stated outside option, without assuming that full elimination or independent negotiation is always preferred.

In the model, a customs union is internally stable if no member obtains a higher government value under a specified outside option than under membership in the customs-union bargaining environment. Let $G_i(0)$ denote the value under continued membership and the common external tariff, and let $G_i(out)$ denote the value under the outside-option benchmark, accounting for the loss of preferential access and the tariff structure applied by the remaining members. In the quantitative exercises below, single-member negotiations and exclusion scenarios are

treated as outside-option benchmarks rather than literal estimates of legal exit. Internal stability requires

$$G_i(0) \geq G_i(out) \quad \text{for all members } i. \quad (8)$$

When this condition holds, exit threats need not imply actual exit. A member may complain about limited flexibility and still remain inside the bloc if the bargaining value of membership dominates the unilateral outside option.

3.2.2 Distributional conflict and external negotiation gridlock

After the participation comparison establishes why a member may remain inside Mercosur, the model turns to a second step: whether the same member governments would approve a proposed external agreement. Conditional on staying in the customs union, each government evaluates the proposed union-wide tariff change against the existing common external policy rather than against exit. This comparison is not only an aggregate welfare calculation, because the common external tariff is a jointly chosen policy that reallocates protection, tariff revenue, and market access across members and sectors. Tariff-revenue rules and member heterogeneity can make governments disagree over the preferred common external tariff (Syropoulos, 2003), and political-economy work shows that production structure and organized interests affect whether customs unions and preferential agreements are attractive (Facchini et al., 2013, 2021). Agreement-formation models make the same point from the external side: customs unions can support or obstruct broader liberalization depending on asymmetry and on the rules governing expansion or collective action (Yi, 1996; Saggi et al., 2013). The approval condition used below therefore asks not whether the bloc gains in aggregate, but whether every member government prefers the proposed tariff change to the status quo.

To evaluate the approval constraint, I use the quantitative environment to trace how each proposed tariff change reallocates gains and losses across sectors before summarizing the result at the member level. Reciprocal liberalization may raise aggregate welfare for the bloc or for some members, but it can impose concentrated losses on politically salient sectors. The protection-for-sale framework explains why organized sectors can receive political weight beyond their aggregate welfare contribution (Grossman and Helpman, 1994). I therefore use the approval margin to ask where those sectoral losses make government support fragile, especially in the larger Mercosur economies, Brazil and Argentina, where external liberalization can expose more diversified manufacturing sectors. This is not a claim that large members always lose from external agreements. It is a claim that small aggregate gains, or modest aggregate losses, can leave the approval margin fragile when the loss side is concentrated.

Once membership is taken as given, external negotiation gridlock occurs when the customs union has a possible union-wide tariff change with an external partner, but the consensus condition is harder to satisfy because at least one member prefers the status quo. Let $G_i(agree)$ denote the value for member i under the union-wide tariff change, and let $G_i(0)$ denote the value under the status quo policy. Approval requires

$$G_i(agree) \geq G_i(0) \quad \text{for all members } i. \quad (9)$$

Member i has a negative approval margin when

$$G_i(0) > G_i(agree). \quad (10)$$

External gridlock occurs when at least one member satisfies this inequality, even if the union remains internally stable and even if smaller members, such as Uruguay, gain from bloc bargaining.

Together, these two conditions separate the forces that keep the customs union together from the forces that prevent liberalization with external partners. The first compares the participation margin, defined as membership in the customs-union bargaining environment relative to a specified outside option. The second compares the proposed tariff change to the status quo from the perspective of each member government. This separation is the theoretical counterpart of the gap emphasized in the introduction: existing customs-union theory explains why unions may

be valuable, fragile, or difficult to expand, while the mechanism here explains why a real customs union can be internally self-enforcing but externally liberalization-blocking. The results section provides quantitative evidence for both margins and relates them to the strategic welfare decomposition and the profit-shifting channel implied by monopolistic competition.

In the empirical sections, I use welfare changes, profit shifts, and sectoral exposure to interpret which members have weaker approval margins. These quantities do not recover lobbying weights, estimate votes, or estimate ratification probabilities.

4 Data and Parameters

The empirical section first defines the data objects and counterfactual construction that discipline the model. The empirical quantities are same-year applied tariffs, sectoral expenditure shares, bilateral trade shares, and sectoral trade elasticities. Because the paper studies a customs union, I also describe how Mercosur’s internal preferences, common external tariff, and exception regimes enter the applied tariff matrix.

Table 1: Empirical quantities used in the counterfactuals

Object	Interpretation
Welfare changes Difference	Real-income percentage changes relative to the same-year baseline. Bloc reciprocal welfare minus own reciprocal welfare. This is a bloc-minus-own-member reciprocal welfare difference, not a full estimate of the value of Mercosur membership.
Selected reciprocal tariff cuts	Tariff cuts selected by the reciprocity rule to minimize the imbalance in concessions.
Remaining imbalance in concessions	Remaining imbalance in concessions after applying the selected tariff cuts.
Zero-tariff benchmark	Full bilateral tariff elimination; it is not a legal forecast.
Sectoral exposure	Labor-share and profit-shift patterns that locate adjustment pressure; these quantities do not estimate lobbying weights, votes, signatures, or ratification probabilities.
Held fixed	Preferences, technology, calibration quantities, iceberg costs, transport costs, non-tariff measures, rules of origin, sectoral elasticities within each calibration, and non-tariff agreement provisions.

4.1 Data

The main data source is GTAP 11 ([Aguilar et al., 2022](#)), which reports bilateral trade, production, consumption, and applied tariff information across countries and sectors. The quantitative sample has 33 sectors and 16 countries or regions: Argentina, Brazil, Paraguay, Uruguay, China, the United States, the European Union, Japan, South Korea, India, Russia, Canada, Mexico, Chile, the rest of South America, and the rest of the world. The rest-of-South-America aggregate includes South American countries not listed separately, many of which have their own regional agreement relationships with Mercosur. Sectors follow the aggregation in [Ossa \(2016\)](#), with the correspondence table reported in the appendix. The benchmark years are 2004, 2007, 2011, 2014, and 2017.

Data coverage. The sectoral and country coverage is important for this application. Some Mercosur studies use richer administrative data for one member or coarser sectoral aggregates for the full agreement environment. [Caliendo and Parro \(2010\)](#), for example, use detailed Uruguayan data but do not model the full set of South American countries, while [Timini and Viani \(2020\)](#) work with a more aggregated sectoral structure. GTAP provides a consistent multi-country, multi-sector baseline that is well suited for comparing members within the

same customs union and for tracing how partner-specific negotiations affect countries with different production structures and protection levels.

4.2 Tariff and trade structure

The applied-tariff matrix and bilateral trade shares determine the initial protection and sourcing patterns used in the counterfactuals. I construct these objects from GTAP effective applied tariffs and trade flows, then use the legal and customs evidence to clarify how national differences in tariff treatment should be interpreted.

Section 2.1 distinguishes common zero rates, positive protection and national departures. Appendix A.6.3 reports the underlying country inventories and customs checks. These recent legal snapshots clarify the interpretation of national differences; they do not reconstruct the model's 2004–2017 tariff matrices.

The retained non-COVID comparisons follow all recovered product codes in eight measures across Argentina, Brazil and Uruguay. No comparable Paraguayan case is completed. Supplier responses differ across these baskets, and their interpretation depends on whether one examines import values or shares. For Uruguay's six-code 2017 basket, Brazil's share falls from 35.56 to 28.56 percent while its sales rise by 1.67 percent. The appendix retains every comparison, including declining and small-value baskets. These episodes describe sourcing around particular measures; they do not estimate the aggregate or causal effect of exemptions.

Figure 1 compares tariffs in both directions for each external partner. Each rate weights the 33 model sectors and four Mercosur members by their bilateral trade in that year. Using the same definition for both sides makes the tariff comparison consistent with the initial bilateral rates in Table 3.

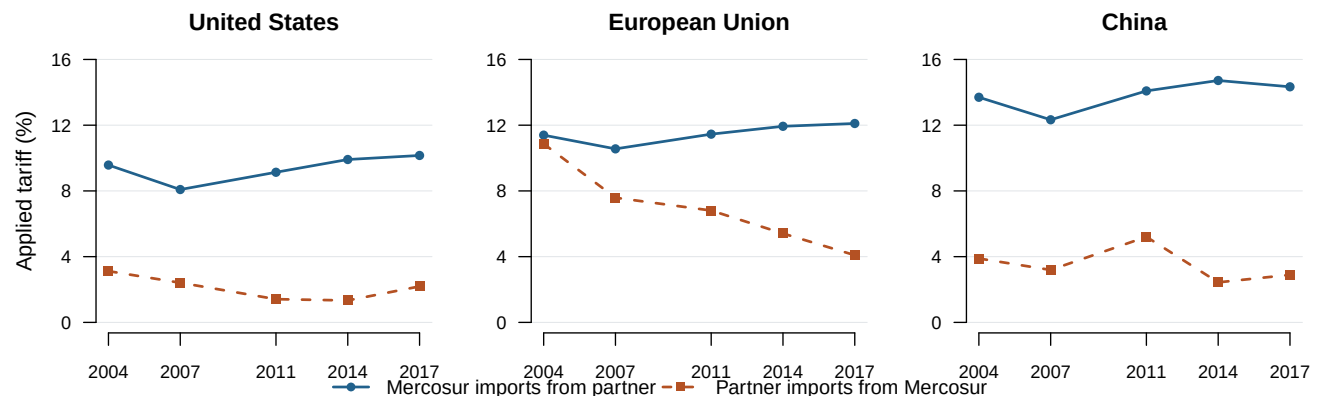


Figure 1: Bilateral applied tariffs between Mercosur and its main external partners, 2004–2017.

Note: GTAP 11 applied rates, weighted by same-year producer-valued bilateral trade across 33 sectors and Argentina, Brazil, Paraguay and Uruguay. Each panel separates tariffs on Mercosur's imports from tariffs on its exports. Weights change across years; the averages do not isolate tariff-policy changes.

Mercosur's trade-weighted import tariffs exceed the corresponding partner rates in these benchmarks. The gap is small for the European Union in 2004: 11.39 percent on Mercosur's imports and 10.86 percent on its exports. The EU rate is particularly high in that benchmark because of agricultural protection.³ These averages describe

³In the saved 2004 matrix, sugar, meat from cattle, sheep, goats and horses, and dairy account for 9.34 percentage points of the 15.51 percent simple average across four exporters and 33 sectors. Giving each cell equal weight magnifies high rates on some small trade flows; weighting by bilateral trade gives 10.86 percent. This meat aggregate alone contributes 4.71 points to that weighted rate. The WTO's contemporaneous review describes a generally open non-agricultural regime alongside substantial agricultural protection ([World Trade Organization, 2004](#)). These are model-sector applied rates. Differences across benchmark years reflect both rates and trade composition and do not identify a discrete policy shock in 2004.

the protection faced by traded goods. They do not measure the extent of legal CET compliance or the economic importance of exemptions.

China became the largest of these three destinations for Mercosur's exports by 2014. In 2017, the bloc exported US\$62.16 billion to China, compared with US\$43.23 billion to the European Union and US\$27.26 billion to the United States. Imports from all three partners were higher in 2017 than in 2004, although they did not rise in every benchmark (Figure 2). These trade patterns provide context for the negotiations; they do not identify the effect of tariff changes.

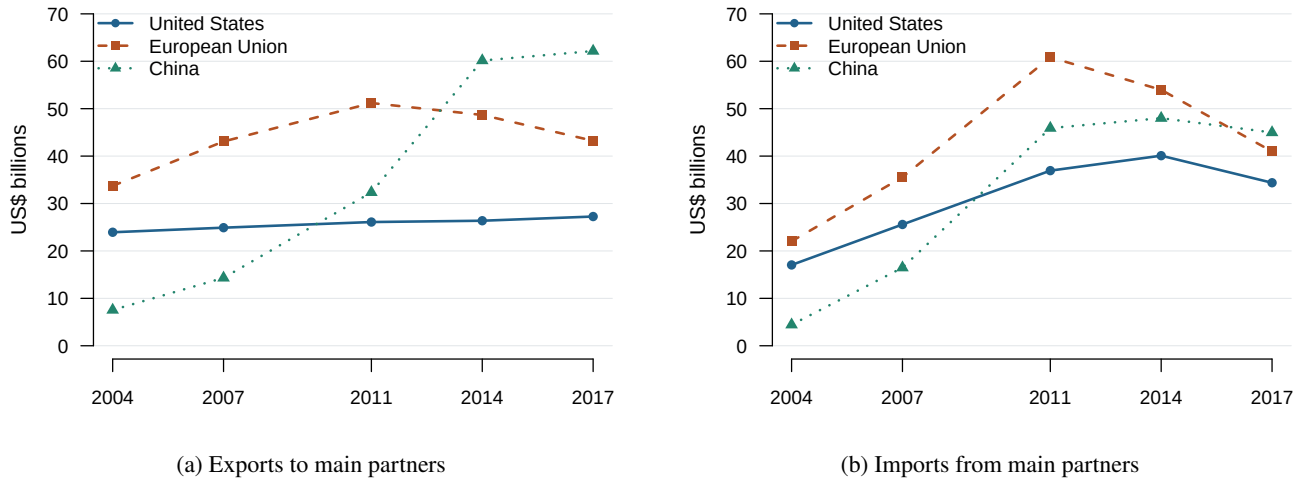


Figure 2: Mercosur trade with the United States, European Union, and China. GTAP 11 producer-valued trade, summed across the 33 model sectors and Argentina, Brazil, Paraguay and Uruguay. Stored values in US\$ millions are divided by 1,000 to obtain billions. Each year retains its reported benchmark units; this is not a constant-price growth series.

4.3 Mapping the theory to the data

The counterfactuals require four empirical quantities: applied bilateral tariffs $t_{ij,s}$, sectoral expenditure shares $\mu_{j,s}$, baseline expenditure shares by origin $\lambda_{ij,s}$, and sectoral trade elasticities. These quantities are sufficient to compute counterfactual equilibria in exact hat algebra.

4.3.1 Tariffs and expenditure shares

I use the existing effective applied-tariff data to define the model's baseline. The legal collection separately documents national tariff heterogeneity; it does not independently verify incorporation of every dated or conditional exception into those rates. Existing exemptions must not be entered again as new cuts, and later statutory gaps cannot be applied to an earlier baseline without reconciliation. Sectoral expenditure shares $\mu_{j,s}$ are computed from GTAP final demand and sum to one within each country. Baseline trade shares $\lambda_{ij,s}$ are constructed from observed bilateral flows and discipline expenditure switching in the counterfactuals.

4.3.2 Counterfactual construction

Across the exercises, counterfactuals are deviations from the same-year matrix of observed effective tariffs and trade shares. I change the tariff vector according to the specified counterfactual and compute the resulting general-equilibrium changes in trade flows, prices, profits, and welfare relative to that benchmark.

The year attached to each exercise identifies the economy used for the calculation. For example, the China 2017

scenario uses 2017 trade and tariff data to compare hypothetical negotiating arrangements. Later official statements explain why these arrangements are relevant to Mercosur’s policy debate. They provide institutional context without supplying the tariff schedules used in the earlier benchmark economies.

That approach is standard in quantitative commercial-policy work (Ossa, 2016; Caliendo and Parro, 2015). In the Mercosur context, Caliendo and Parro (2010) apply a multi-sector Ricardian model to Uruguay’s partner-choice problem, comparing continued Mercosur participation with a United States agreement and a leaving-Mercosur counterfactual. Timini and Viani (2020) use structural gravity to study the EU–Mercosur agreement by mapping estimated trade-flow effects into general-equilibrium trade-cost changes. I use the same counterfactual logic but compare bloc bargaining, member bargaining, zero-tariff benchmarks, and exclusion cases.

The main exercises use the observed applied-tariff matrix so that each comparison starts from the same country-year policy environment. I define each counterfactual by the partner, tariff change and negotiating party, and specify separately whether a single-member deviation loses intrabloc preferences.

I keep the counterfactual design narrow to avoid confusion over the scope of the study, with a focus on tariff structures. Tariff counterfactuals change the bargaining side and the specified bilateral tariff schedules while holding preferences, technology, calibration quantities, and the non-tariff environment fixed. Exclusion and lost-preference cases are separate outside-option benchmarks; they are not estimates of a dynamic legal withdrawal, ratification process, or implementation path.

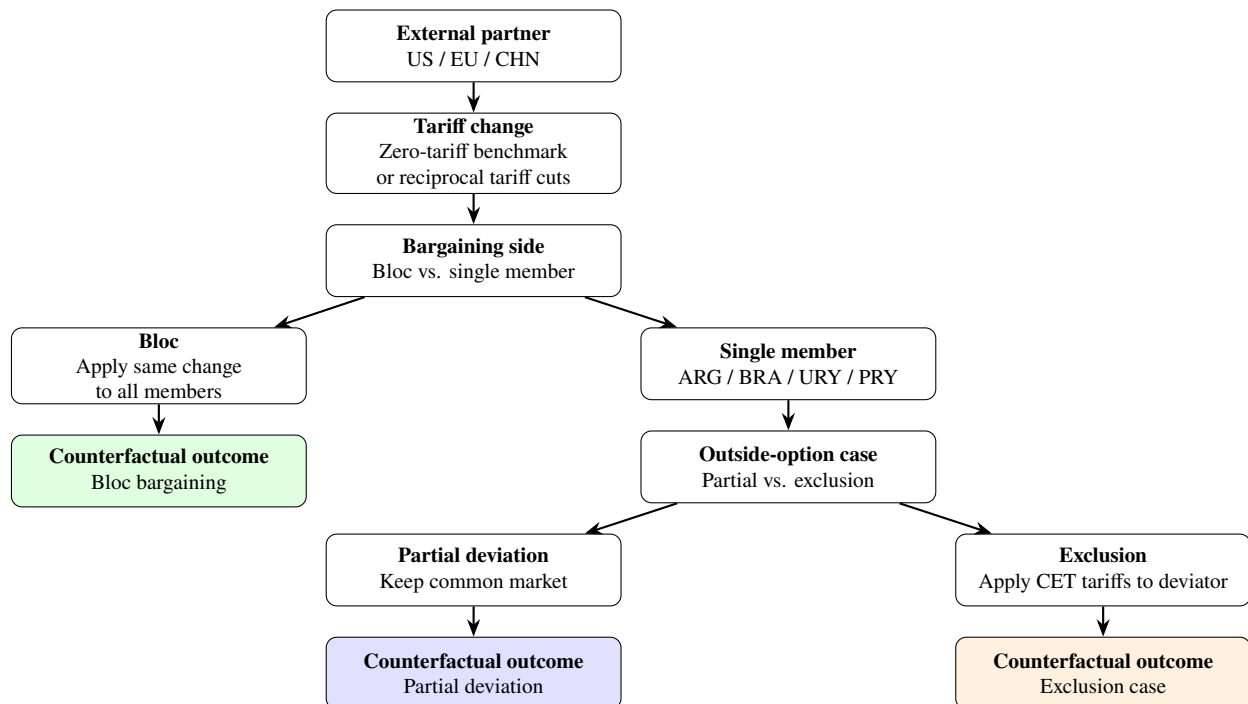


Figure 3: Economic classification of counterfactual cases

For each member–partner pair, I construct two tariff changes. The zero-tariff benchmark sets the affected bilateral tariffs to zero, while the reciprocal tariff-cut counterfactual uses the concession-balance criterion in Section 3.1.1 to select the candidate tariff cuts that minimize the imbalance under the real-net-export concession measure. Bagwell and Staiger (1999) provide the terms-of-trade rationale for reciprocal trade agreements, and Bagwell and Staiger (2016) survey how trade-agreement design disciplines tariff choices. Beshkar et al. (2025) motivates comparing concessions, with the differences in the operational metric stated in Section 3.1.1, while Beshkar and Lee (2022) explains how market power matters for trade-agreement negotiations. Bagwell et al. (2021) provides a related quantitative benchmark for studying multiparty tariff negotiations. When a deviation triggers exclusion, remaining

members treat the deviator as a nonmember and apply their effective external tariffs to imports from the deviator.

4.3.3 Trade elasticities

Sectoral elasticities are calibrated from WITS/TRAINS-BACI tariff and trade data mapped into the 33-sector model aggregation. I use WITS rather than estimating the vector directly from the GTAP model database because the GTAP database is already aggregated; at that level, the triple-ratio equation identifies only a small number of sectors directly. WITS keeps more product and reporter-partner variation before aggregation, which makes the elasticity calculation more transparent and increases the precision of the estimates. The dependent variable uses BACI trade values without adding tariff payments, represented by producer-valued flows $x_{ij,s}$. Under this convention, the gravity-style moment condition is

$$\ln \frac{x_{ij,s}x_{jk,s}x_{ki,s}}{x_{ji,s}x_{kj,s}x_{ik,s}} = -\sigma_s \ln \frac{(1+t_{ij,s})(1+t_{jk,s})(1+t_{ki,s})}{(1+t_{ji,s})(1+t_{kj,s})(1+t_{ik,s})} + \epsilon_{ijk,s}. \quad (11)$$

If tariff-inclusive expenditure $X_{ij,s} = (1+t_{ij,s})x_{ij,s}$ were used instead, the tariff coefficient would be $1 - \sigma_s$. Thus the negative of the estimated producer-value coefficient maps directly to the CES substitution elasticity σ_s ; no additional unit is added to the estimated or fallback calibration values. For the main calibration, I estimate a single WITS vector by pooling the five benchmark years, 2004, 2007, 2011, 2014, and 2017. When a sector estimate is not usable, I retain the benchmark elasticity vector from [Ossa \(2016\)](#). The appendix reports the full sector table, the WITS year-specific robustness vectors, the GTAP direct-estimation check, and the Ossa benchmark comparison. Across the four focal calibrations, Argentina, Brazil, and Uruguay retain the signs of reciprocal bloc welfare, own-member welfare, and their difference in China 2017 and United States 2004, although magnitudes vary. Paraguay retains these signs in United States 2004 but not in China 2017.

5 Results

The counterfactuals quantify the stability-with-gridlock mechanism by showing that participation and approval move differently across members. The participation margin asks whether members gain more from bargaining through Mercosur than from bargaining alone. The approval margin asks whether the same bloc counterfactual is attractive member by member once aggregate welfare is read with sectoral exposure. The evidence consists of welfare changes, sectoral exposure, and concentrated losses. I do not estimate signature, ratification, or lobbying probabilities; doing so would require additional assumptions about political preferences. All focal welfare values in this section use the WITS-main calibration.

5.1 Counterfactual comparisons

The participation comparison places bloc and member bargaining under the same tariff baseline and reciprocity rule. For each partner-year, I keep the same observed applied tariffs and trade shares and construct four tariff counterfactuals: Mercosur bloc reciprocal tariff cuts, single-member reciprocal tariff cuts, a Mercosur bloc zero-tariff case, and a single-member zero-tariff case. The comparison therefore asks whether the same external opportunity is more valuable as a bloc counterfactual or as a member's outside option. Across these comparisons, the bargaining side and tariff schedule change, while the calibration and non-tariff environment are held fixed.

That comparison is closest to [Caliendo and Parro \(2010\)](#), but it changes the question. They study Uruguay's possible agreement with the United States and find positive welfare effects for Uruguay from bilateral tariff reductions, with most gains coming from access to lower-cost intermediate inputs; when they eliminate tariffs between Mercosur and the United States, Uruguay has the largest gains while Argentina and Brazil benefit little. I keep the Uruguay–United States episode, but separate the bloc counterfactual from Uruguay's own counterfactual

under the same reciprocal tariff rule. I use the same distinction for Uruguay–China, where the key comparison is between Uruguay’s payoff through Mercosur and its payoff when bargaining alone under the same reciprocal rule. That is why the later United States 2004 result compares Uruguay’s 0.867 percent bloc reciprocal gain with its -0.006 percent own reciprocal outcome under the main WITS calibration.

Reciprocal cases are counterfactual tariff cuts selected by the reciprocity rule defined in Section 3.1.1. The zero-tariff case sets the affected bilateral tariffs to zero and measures the welfare scale of full bilateral tariff elimination; it is not a legal prediction under Mercosur rules.

Table 2: Counterfactual tariff comparisons

Label	Object	Interpretation
Bloc reciprocal	Mercosur negotiates jointly with the partner	Main approval and bargaining object under net-market-access reciprocity
Single-member reciprocal	One member negotiates alone with the partner	Flexibility benchmark under the same reciprocity rule
Bloc zero-tariff FTA benchmark	Mercosur and the partner eliminate bilateral tariffs	Full-liberalization benchmark for the bloc
Small-country zero-tariff FTA benchmark	One member and the partner eliminate bilateral tariffs	Outside-option benchmark for unilateral external flexibility

Notes: Reciprocal exercises use selected reciprocal tariff cuts; zero-tariff exercises are benchmark liberalizations. Welfare changes are percentage changes in real income relative to the same-year applied-tariff baseline; differences between these percentage changes are reported in percentage points.

The selected reciprocal cuts come from a restricted positive boundary grid over proportional tariff reductions, and Table 3 reports the resulting trade-weighted average tariff cuts⁴. For the China and United States partner-year rows reported in Table 3, the selected tariff cuts minimize the remaining imbalance in concessions over the restricted one-percentage-point search set. The same restricted search set is used for the single-member reciprocal benchmarks in the participation comparison. Table 3 reports the initial average tariff and the implied average reciprocal tariff cut on each side, rather than the proportional grid choice. Reporting average tariff cuts is useful because the model is evaluated under a concrete trade-weighted tariff schedule, not under verbal labels such as “large cut” or “small cut.”

⁴Only for the purpose of this table.

Table 3: Initial tariffs and average reciprocal tariff cuts

Partner	Year	Mercosur imports		Partner imports	
		Initial tariff	Avg. cut	Initial tariff	Avg. cut
China	2004	13.70	6.44	3.89	3.89
China	2007	12.33	2.34	3.19	3.19
China	2011	14.08	5.07	5.21	5.21
China	2014	14.72	2.65	2.44	2.44
China	2017	14.34	3.44	2.89	2.89
United States	2004	9.57	9.57	3.12	2.99
United States	2007	8.08	3.80	2.41	2.41
United States	2011	9.13	1.28	1.41	1.41
United States	2014	9.91	0.99	1.34	1.34
United States	2017	10.16	2.23	2.19	2.19

Notes: Initial tariffs use baseline import weights over affected bilateral sector flows and are reported in percent; average cuts are in percentage points. The average cut is the difference between the initial average tariff and the counterfactual average tariff implied by the selected proportional cut. These schedules were selected under the original SIGMA calibration and are held fixed in the WITS evaluation. Mercosur imports are tariffs imposed by Mercosur members on the partner; partner imports are tariffs imposed by the partner on Mercosur exporters.

The China 2017 exercise starts from an average Mercosur tariff of 14.34 percent on imports from China and an average Chinese tariff of 2.89 percent on imports from Mercosur. The selected cuts are 3.44 and 2.89 percentage points, respectively. In the United States 2004 exercise, Mercosur's initial average tariff is 9.57 percent and its selected cut is 9.57 percentage points; the corresponding United States values are 3.12 percent and 2.99 percentage points. These schedules minimize the measured imbalance in concessions over the specified candidate set, providing a concrete comparison of tariff exchanges starting from asymmetric protection ([Bagwell and Staiger, 1999, 2016](#); [Beshkar et al., 2025](#)).

Every selected reciprocal scenario eliminates one side's affected tariffs because the search requires at least one side to do so. The zero-tariff benchmark eliminates both sides' affected tariffs and measures the consequences of full bilateral liberalization. It provides a reference for interpreting the reciprocal scenarios. Actual agreements can distribute concessions across implementation periods, sensitive products and commitments beyond tariffs, so their acceptability depends on features outside this comparison.

A common scale helps put these selected tariff cuts in perspective: I compare them with the tariff schedule negotiated in the EU–Mercosur trade agreement. Annex 2-A is the agreement's product-level tariff schedule: it records the base tariff and staging category used to phase down tariffs for each covered line. Aggregating product-level post-agreement tariffs in the 2022 coded schedule with the existing trade weights, the initial average tariff is 2.34 percent on EU imports from Argentina and Brazil and 4.95 percent on their imports from the EU. The corresponding post-agreement averages are 1.59 and 1.24 percent, so the average cuts are 0.75 and 3.71 percentage points, or 32.0 and 74.9 percent of the initial trade-weighted tariffs. These averages cover Argentina and Brazil rather than all four Mercosur members and include positive-value services flows in the denominator. The comparison is contextual because the coverage and models differ; the EU–Mercosur agreement itself is not the counterfactual being evaluated here.

5.2 Participation Margin

To evaluate the participation margin, I hold the partner, year, tariff baseline, and reciprocity rule fixed and compare each member's welfare under Mercosur's reciprocal tariff cuts with its welfare when bargaining alone. Table 4 reports that comparison and the bloc-minus-own-member difference. A positive difference means that participating

through Mercosur raises a member's payoff relative to the independent negotiation. A negative value means the independent route is stronger. Krueger (1997) gives the institutional reason why this comparison matters: free trade agreements preserve national external tariffs, while customs unions require common external policy. Lake (2019) turns that distinction into a dynamic flexibility argument, in which FTA members can form future agreements independently but customs-union members must move jointly. Lake and Yildiz (2016) complements this by explaining why customs unions tend to be geographically concentrated. In the model here, that flexibility benefit is weighed against the bargaining value of acting through Mercosur.

The general participation result, therefore, is that small members may value the bloc not because autonomy is worthless, but because reciprocity changes the tariff cuts they can obtain alone. This is a stricter comparison than the zero-tariff outside option. External partners begin from relatively low tariffs, so full tariff elimination abroad has limited room to improve Uruguay's market access; the reciprocal exercise asks whether Uruguay can match the bloc's reciprocal outcome on its own.

Table 4: Participation payoffs under reciprocal and zero-tariff cases

Scenario	Country	Bloc	Own member	Difference
<i>Panel A. Reciprocal tariff bargaining</i>				
China 2017	Argentina	0.094	0.102	-0.007
China 2017	Brazil	0.131	0.129	0.002
China 2017	Uruguay	0.598	-0.232	0.830
United States 2004	Argentina	-0.077	0.062	-0.138
United States 2004	Brazil	-0.111	0.047	-0.158
United States 2004	Uruguay	0.867	-0.006	0.874
<i>Panel B. Zero-tariff outside-option benchmark</i>				
China 2017	Argentina	-0.125	-0.105	-0.020
China 2017	Brazil	-0.147	-0.126	-0.020
China 2017	Uruguay	0.303	0.337	-0.034
United States 2004	Argentina	-0.070	-0.035	-0.035
United States 2004	Brazil	-0.105	-0.070	-0.035
United States 2004	Uruguay	0.950	0.948	0.002

Notes: Bloc and own-member entries are percentage changes in real income relative to the same-year benchmark equilibrium with observed applied tariffs and trade shares. Panel A uses the selected reciprocal tariff cuts. Panel B sets bilateral tariffs to zero as a zero-tariff outside option. The Difference equals bloc welfare minus own single-member welfare, in percentage points.

Uruguay is where the gains are largest from bloc bargaining in the China and United States reciprocal counterfactuals. In the China 2017 exercise, Uruguay gains 0.598 percent under bloc reciprocal bargaining and loses 0.232 percent in its own single-member reciprocal exercise, yielding a participation difference of 0.830 percentage points. In the United States 2004 exercise, Uruguay gains 0.867 percent under bloc reciprocal bargaining and is essentially unchanged when bargaining alone, with a -0.006 percent own-member effect and a 0.874 percentage-point difference. This is the participation-margin result: the flexible outside option exists, but under the reciprocal tariff-cut rule Uruguay does not obtain alone the reciprocal outcome it receives through the bloc.

The zero-tariff exercises compare full bilateral liberalization negotiated by the bloc with the same tariff treatment negotiated by one member. Both sides remove the tariffs covered by the agreement. The flexibility argument in Lake (2019) and Lake and Yildiz (2016) concerns a member's ability to negotiate an external agreement without waiting for the bloc. Panel B of Table 4 evaluates that option for the specified zero-tariff schedules. Full elimination is a benchmark, not a legal forecast or a welfare upper bound.

Panel B presents a welfare comparison. It asks whether a member gains relative to its own baseline equilibrium under full bilateral liberalization. In 2004, Uruguay's own zero-tariff case with the United States yields a 0.948 percent gain, almost the same as the 0.950 gain under the bloc case. In the China 2017 zero-tariff case, Uruguay's

own gain is 0.337 percent, compared with 0.303 under the bloc case. These figures explain why the zero-tariff outside option can look attractive for Uruguay when compared to the country reciprocal negotiation, but they do not compare aggregate economic size. Their relevance is to compare specified tariff schedules motivated by the historical negotiating alternatives.

5.3 Approval Margin

To evaluate approval, I use the same bloc counterfactuals to ask whether each member government has a country-level case for supporting the agreement. I start with member-level welfare because the agreement must be attractive country by country, not only for the bloc as a label. Panel A of Table 4 reports the two focal reciprocal cases that carry the main interpretation: China 2017, where Mercosur's trade relationship with China is mature in the sample, and United States 2004, which speaks to the Uruguay–United States negotiation episode. Welfare is reported to three decimals because the large-member magnitudes are small. Appendix A.7 reports the same focal welfare results under the Ossa benchmark vector, direct GTAP pooled check, and WITS early-year robustness. Argentina, Brazil, and Uruguay retain the signs of reciprocal bloc welfare, own-member welfare, and their difference across these focal calibrations, although magnitudes vary. Paraguay's China 2017 signs are not stable across them.

Brazil and Argentina face a different approval margin from Uruguay. In China 2017, both large members gain under bloc reciprocal bargaining, but the gains are small in relative terms: 0.131 percent for Brazil and 0.094 for Argentina, compared with 0.598 for Uruguay. In United States 2004, both large members lose in aggregate under the bloc reciprocal exercise, with -0.111 percent for Brazil and -0.077 for Argentina, while Uruguay gains 0.867 percent. These estimates give the numerical counterpart to the paper's stability-with-gridlock mechanism: the same reciprocal tariff cuts that strengthen Uruguay's participation margin can make the large-member approval margin fragile. This pattern is consistent with several mechanisms in the customs-union literature. For instance, [Syropoulos \(2003\)](#) shows that tariff-revenue rules and member heterogeneity can change members' preferred common external tariff.⁵ The result is also consistent with [Campos and Timini \(2022\)](#), who emphasize unequal Mercosur gains across countries and groups.

The participation comparison leaves the distribution of the bloc scenario's effects across members to be assessed separately. In China 2017, Brazil and Argentina have positive but small aggregate gains; in United States 2004, both have negative aggregate effects while Uruguay gains. Terms-of-trade changes, operating profits and tariff revenue can contribute differently to these outcomes. The sectoral evidence below identifies where the selected reciprocal scenarios concentrate losses.

The small aggregate gains found for Brazil and Argentina in some cases leave open the question of which industries gain and lose from liberalization. I use sectoral exposure to identify where the approval constraint is likely to be politically fragile. This subsection links the model welfare decomposition to the approval margin. The quantitative model generates welfare changes, profit shifts, and labor-share adjustment; the political-economy question is where those effects are concentrated. I use labor-share movements as sectoral exposure: they identify the sectors in which workers and firms absorb the adjustment, but they are not estimates of lobbying weights. This is the bridge to the Grossman–Helpman approval logic, where organized or politically sensitive sectors can matter more than their aggregate welfare share ([Grossman and Helpman, 1994, 1995](#)).

Differences in specialization can help explain why industry gains vary across countries. Countries with more concentration of industries in the export-competing sectors might lose less in some industries when liberalizing. Table 5 makes that bridge relative to each country's own baseline structure. Uruguay is smaller and more specialized than Brazil, so raw cross-country concentration would be misleading. The baseline effective number of sectors summarizes the diversification of the initial labor allocation; the exposure effective number summarizes the

⁵An important note in the argument here is also presented for Customs Union studies, [Olarreaga et al. \(1999\)](#) show that the common external tariff reflected both economic incentives and political-economy pressures, and [Saggi et al. \(2013\)](#) shows that customs unions can support or obstruct broader liberalization depending on institutional conditions.

concentration of the gain- or loss-side labor-share movement. The ratio is the baseline effective number divided by the exposure effective number, so values above one mean that the counterfactual exposure is more concentrated than the country's own baseline economy.

Table 5: Approval-margin exposure, baseline specialization, and concentrated exposure

Scenario	Country	Side	Welfare	Realloc.	Base EffN	Shock EffN	Ratio	Top sector	Top share	Base share
China 2017	Brazil	Gain	0.131	0.40	16.68	3.94	4.23	Meat: cattle, sheep, goats, horse	33.8	4.8
China 2017	Brazil	Loss	0.131	0.40	16.68	6.21	2.68	Mineral products nec	29.4	9.6
China 2017	Uruguay	Gain	0.598	3.04	15.03	1.19	12.62	Meat: cattle, sheep, goats, horse	91.5	12.4
China 2017	Uruguay	Loss	0.598	3.04	15.03	10.81	1.39	Chemical, rubber, plastic prods	18.9	8.3
United States 2004	Brazil	Gain	-0.111	0.85	15.33	3.56	4.31	Leather products	47.9	2.5
United States 2004	Brazil	Loss	-0.111	0.85	15.33	2.53	6.07	Chemical, rubber, plastic prods	59.8	18.4
United States 2004	Uruguay	Gain	0.867	7.22	15.60	1.51	10.33	Meat: cattle, sheep, goats, horse	78.5	12.9
United States 2004	Uruguay	Loss	0.867	7.22	15.60	8.00	1.95	Chemical, rubber, plastic prods	26.4	10.9

Notes: Focal reciprocal bloc cases, with welfare and exposure both evaluated under the WITS-main calibration. Argentina and Paraguay are reported in Appendix A.4. Welfare is the reciprocal bloc real-income change in percent. Realloc. is the total absolute labor-share movement on the gain or loss side, in percentage points. Base EffN and Exposure EffN are inverse-Herfindahl effective numbers of sectors for the baseline labor allocation and the exposure-side distribution. Ratio is Base EffN divided by Exposure EffN. Top share and Base share are percentages. Appendix A.3 gives formulas and top-sector drivers.

One implication is that Uruguay's gain-side concentration is not just a mechanical consequence of being smaller. In United States 2004, Uruguay's gain-side exposure has an effective number of 1.51 sectors, compared with a baseline effective number of 15.60. Meat alone accounts for 78.5 percent of the gain-side movement, even though it is 12.9 percent of Uruguay's baseline labor allocation, and dairy accounts for the remainder. In China 2017, Uruguay's gain-side exposure has an effective number of 1.19 sectors, compared with a baseline effective number of 15.03; meat accounts for 91.5 percent of the gain-side movement, with textiles, wool, and oil seeds next. This is why the small-member participation result can be politically legible: the bloc bargain creates gains in sectors that are already central to Uruguay's export structure, but the gain-side exposure is still more concentrated than the baseline economy.

A second implication is that Brazil's approval margin differs across partners. In United States 2004, Brazil's aggregate welfare effect is negative, -0.111 percent, and the loss-side exposure is concentrated relative to its baseline economy: the loss-side effective number is 2.53, compared with a baseline effective number of 15.33. Chemical, rubber, and plastic products alone account for 59.8 percent of the loss-side movement and 18.4 percent of baseline labor. In China 2017, the claim is different. Brazil's aggregate welfare effect is positive but small, 0.131 percent, and the loss-side exposure is still more concentrated than the baseline allocation: the loss-side effective number is 6.21 against a baseline effective number of 16.68. The weak approval margin comes from manufacturing exposure relative to a small aggregate gain, not one dominant losing sector. The top five losing sectors—mineral products, wearing apparel, other manufactures, chemical, rubber, and plastic products, and electronic equipment—account for 78.0 percent of the loss-side movement.

Numerically, this is the counterpart to the labor-market and China-exposure mechanisms in [Dix-Carneiro \(2014\)](#), [Dix-Carneiro and Kovak \(2017\)](#), [Costa et al. \(2016\)](#), and [Paz \(2018\)](#). I do not estimate labor-market dynamics here. The interpretation is narrower: the same reciprocal tariff cuts can give Uruguay concentrated, politically legible gains while giving Brazil either concentrated losses or diffuse manufacturing losses that are large relative to a small aggregate payoff. That is the approval-margin channel.

Taken together, the results separate the participation and approval margins in model units. In the focal reciprocal

tariff-cut counterfactuals, bloc reciprocity raises Uruguay's payoff relative to its own reciprocal benchmark, which supports the bargaining-power component of the participation argument. The same counterfactuals can generate small or negative aggregate payoffs for Brazil and Argentina, and sectoral exposure identifies weak approval margins. Paraguay's appendix diagnostics show additional variation across scenarios. Geography and transit costs are outside the model, so the exercise does not attribute Paraguay's participation incentives to those channels. Stability and external-agreement gridlock are therefore not contradictory outcomes in the model; they are different margins of the same negotiation problem.

6 Policy Discussion and Framework Application

The counterfactuals separate Mercosur's external-negotiation problem into three issues: the value of joint bargaining, the value of national flexibility, and the welfare changes, sectoral exposure, and concentrated losses created by external tariff changes. The policy lesson is conditional: given its joint-negotiation institutions and tariff structure, Mercosur can be stable on the participation margin and fragile on the approval margin when negotiating with large external partners.

The selected tariff scenarios make the implications of asymmetric initial protection explicit. A partner can eliminate a large proportion of already-low tariffs while Mercosur makes a smaller proportional cut in higher tariffs. I compare the resulting concessions using the model's measure of the fall in real net exports relative to the baseline. Member welfare and sectoral exposure then indicate which economies gain from the selected exchange and where losses are concentrated. This sequence connects market-access concessions to participation and approval, while leaving acceptance of a complete agreement to broader economic and political considerations.

In the calibrated exercises, bloc reciprocal bargaining is especially valuable for the smaller member. Uruguay receives larger gains when it negotiates through Mercosur than when it negotiates alone under the reciprocal criterion, especially in the China exercises, as predicted by the theory of Customs Union. This supports the bargaining-power component of the participation argument: even when Uruguay demands more external flexibility, the customs union can still provide bargaining value.

The zero-tariff benchmark asks what full bilateral tariff elimination would imply under either bloc or single-member negotiation. Both sides remove the affected tariffs, without selecting cuts through the concession measure. Comparing the two negotiating arrangements under this benchmark shows how the value of collective bargaining depends on the extent of liberalization. The Uruguay–United States case shows that full bilateral elimination can be attractive for Uruguay under the specified baseline; it does not imply that Uruguay can obtain partner concessions by cutting its own tariffs alone.

That result does not contradict the reciprocal result; it clarifies the policy trade-off. Uruguay's modeled outside-option value is economically meaningful, but the ranking of options depends on the tariff counterfactual. Under reciprocal bargaining, Mercosur can improve Uruguay's position by pooling market access. Under the zero-tariff benchmark, national flexibility can become more attractive because the member receives direct partner liberalization without solving the bloc bargaining problem. This comparison also depends on the benchmark year.

For Brazil and Argentina, the relevant issue is less the value of remaining in Mercosur and more the domestic approval of a new external agreement. The large members have more diversified industrial structures and larger import-competing sectors. A reciprocal agreement can therefore generate small or negative aggregate effects while concentrating losses in politically salient sectors. This is the approval-margin exposure: the agreement may be attractive for the small member and fragile for the large member whose losses are concentrated.

Existing tariff flexibility affects members on both sides of a transaction. An external cut may lower an importer's costs while narrowing a regional exporter's preference. The supplier comparisons make this channel relevant to examine, but do not isolate its causal size. For qualifying intrabloc goods that already enter at zero, a change in sales after an external cut does not imply a reduction in their own tariff.

The direction of an exception also matters for agreement design. A lower rate can increase import competition, whereas an exception preserving protection can reduce it. Neither is interchangeable with a transfer compensating a losing sector. Whether exceptions, phase-ins or compensation strengthen participation or ease approval depends on the affected sectors and the proposed package. The model measures welfare and sectoral exposure under specified tariff scenarios; it does not estimate whether an exemption or transfer removes a political veto.

The model maps tariff changes into welfare changes and sectoral exposure, and then uses those margins to interpret Mercosur's combination of persistent membership and slow external liberalization. The results are consistent with a bloc that is valuable enough to hold together but internally heterogeneous enough to make approval of large external agreements more fragile. They should not be read as an estimate of lobbying contributions, legislative votes, or the probability that any specific agreement will be signed.

7 Conclusion

The evidence points to a specific form of customs-union stability: Mercosur can remain valuable as a bargaining platform even when member-by-member approval of external agreements is fragile. The central distinction is between participation and approval. Participation compares the value of staying in the customs union with unilateral outside options, including the loss of preferences when exit or exclusion is modeled. Approval compares a proposed bloc agreement with the status quo for each member government. The quantitative exercises show that these margins can move in different directions.

Reciprocal bargaining results show that Mercosur provides bargaining value to the small member. Uruguay gains more from negotiating through the bloc than from negotiating alone in the main reciprocal exercises, especially with China. Exit threats and demands for flexibility therefore need not imply that the reciprocal outside option dominates the bloc bargain. The customs union can remain valuable because it pools market access and improves the bargaining position of a small economy.

Approval results explain why stability does not imply external liberalization. Brazil and Argentina often receive small or negative aggregate effects from reciprocal bloc agreements, and the associated losses can be concentrated in politically salient sectors. Uruguay receives larger gains in the key reciprocal cases, with gains concentrated in sectors that make the agreement easier to defend domestically. The same bloc agreement can therefore look attractive from the perspective of small-member bargaining power and fragile from the perspective of large-member approval.

The zero-tariff results show why the extent of liberalization and the negotiating party must be considered together. Full bilateral elimination can be valuable for Uruguay, especially in the United States 2004 case. Comparing bloc and single-member agreements under both full elimination and selected reciprocal cuts shows why a demand for flexibility need not imply a preference for leaving the customs union under every agreement design.

The institutional evidence explains why these comparisons remain relevant when applied tariffs differ across members. The country inventories document national relief alongside positive common protection, while the negotiation documents show an exchange of concessions. The inventories do not measure the bloc-wide share of production protected or the causal effect of exemptions on trade (Appendix A.6.3). The model instead evaluates specified agreements from observed applied-tariff baselines, asking how joint bargaining and the resulting distribution of gains affect participation and approval.

Across the tariff scenarios studied here, collective bargaining can retain value for members while a particular external offer generates weak approval incentives. The common external policy pools market access, and members evaluate the resulting exchange against different domestic gains and losses. The counterfactuals identify economic inputs into that decision: welfare changes and sectoral exposure under specified tariff schedules. Explaining acceptance of a complete agreement would also require evidence on political preferences, compensation and commitments beyond tariffs. The contribution is to show how membership value and approval incentives can differ

within the same customs union.

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Appendix

A.1 Quantitative Equilibrium Inputs and Aggregation

The counterfactuals use an Ossa-style multi-sector CES environment solved in exact hat algebra. For each benchmark year, the required quantities are applied bilateral tariffs $t_{ij,s}$, baseline expenditure shares by origin $\lambda_{ij,s}$, sectoral expenditure shares $\mu_{j,s}$, and sectoral elasticities σ_s . A counterfactual tariff vector $\tilde{\mathbf{t}}$ changes delivered prices and expenditure shares; the solved equilibrium returns changes in welfare, sectoral profits, trade flows, and labor-share exposure relative to the same-year baseline. These quantities support the participation and approval margins.

Table 6: Quantitative sample, data, and aggregation

Object	Aggregation used in the quantitative exercises																																	
Countries and regions	Sixteen model regions: Argentina, Brazil, Paraguay, Uruguay, China, United States, European Union, Japan, South Korea, India, Russia, Canada, Mexico, Chile, rest of South America, and rest of world.																																	
Mercosur bloc	Argentina, Brazil, Paraguay, and Uruguay. Bloc counterfactuals change the bilateral tariffs between this set and the partner; member-level welfare, profit, and labor-share outcomes are then read country by country.																																	
Sectors	Thirty-three GTAP/Ossa sectors. The compact correspondence is: <table><tr><td>1 Raw and pro-processed rice</td><td>12 Wool, silk-worm cocoons</td><td>23 Paper products, publishing</td></tr><tr><td>2 Wheat</td><td>13 Forestry</td><td>24 Chemical, rubber, plastic prods</td></tr><tr><td>3 Cereal grains nec</td><td>14 Meat: cattle, sheep, goats, horse</td><td>25 Mineral products nec</td></tr><tr><td>4 Vegetables, fruit, nuts</td><td>15 Meat products nec</td><td>26 Ferrous metals</td></tr><tr><td>5 Oil seeds</td><td>16 Vegetable oils and fats</td><td>27 Metals nec</td></tr><tr><td>6 Raw and pro-processed sugar</td><td>17 Food products nec</td><td>28 Metal products</td></tr><tr><td>7 Plant-based fibers</td><td>18 Beverages and tobacco products</td><td>29 Motor vehicles and parts</td></tr><tr><td>8 Crops nec</td><td>19 Textiles</td><td>30 Transport equipment nec</td></tr><tr><td>9 Cattle, sheep, goats, horses</td><td>20 Wearing apparel</td><td>31 Electronic equipment</td></tr><tr><td>10 Animal products nec</td><td>21 Leather products</td><td>32 Machinery and equipment nec</td></tr><tr><td>11 Raw and pro-processed dairy</td><td>22 Wood products</td><td>33 Manufactures nec</td></tr></table>	1 Raw and pro-processed rice	12 Wool, silk-worm cocoons	23 Paper products, publishing	2 Wheat	13 Forestry	24 Chemical, rubber, plastic prods	3 Cereal grains nec	14 Meat: cattle, sheep, goats, horse	25 Mineral products nec	4 Vegetables, fruit, nuts	15 Meat products nec	26 Ferrous metals	5 Oil seeds	16 Vegetable oils and fats	27 Metals nec	6 Raw and pro-processed sugar	17 Food products nec	28 Metal products	7 Plant-based fibers	18 Beverages and tobacco products	29 Motor vehicles and parts	8 Crops nec	19 Textiles	30 Transport equipment nec	9 Cattle, sheep, goats, horses	20 Wearing apparel	31 Electronic equipment	10 Animal products nec	21 Leather products	32 Machinery and equipment nec	11 Raw and pro-processed dairy	22 Wood products	33 Manufactures nec
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10 Animal products nec	21 Leather products	32 Machinery and equipment nec																																
11 Raw and pro-processed dairy	22 Wood products	33 Manufactures nec																																
Benchmark years	2004, 2007, 2011, 2014, and 2017. Each counterfactual starts from the same-year applied tariff matrix, expenditure shares, trade shares, and elasticity vector.																																	
Policy objects	Bloc reciprocal, single-member reciprocal, bloc zero-tariff benchmark, and small-country zero-tariff benchmark. Reciprocal exercises choose proportional tariff cuts from the maximal-concession frontier that best balance real net-market-access changes between the bargaining sides.																																	

Table 7: Selected reciprocal tariff cuts

Partner	Year	Mercosur imports		Partner imports	
		Initial tariff	Avg. cut	Initial tariff	Avg. cut
China	2004	13.70	6.44	3.89	3.89
China	2007	12.33	2.34	3.19	3.19
China	2011	14.08	5.07	5.21	5.21
China	2014	14.72	2.65	2.44	2.44
China	2017	14.34	3.44	2.89	2.89
United States	2004	9.57	9.57	3.12	2.99
United States	2007	8.08	3.80	2.41	2.41
United States	2011	9.13	1.28	1.41	1.41
United States	2014	9.91	0.99	1.34	1.34
United States	2017	10.16	2.23	2.19	2.19

Notes: Initial tariffs are baseline-import-weighted averages over affected bilateral sector flows, in percent; average cuts are in percentage points. The average cut is the difference between the initial average tariff and the counterfactual average tariff implied by the selected proportional cut. These schedules were selected under the original SIGMA calibration and are held fixed in the WITS evaluation. Mercosur-side averages refer to tariffs imposed by Mercosur members on imports from the partner; partner-side averages refer to tariffs imposed by the partner on imports from Mercosur exporters.

A.2 Baseline Trade Structure and Intra-bloc Exposure

The main text reports counterfactual changes because the participation and approval margins are defined relative to the same-year applied-tariff status quo. Table 8 adds the corresponding baseline levels. This is useful for interpreting why percentage welfare changes are larger for small members and why low blocwide intrabloc trade can still matter for the participation margin of individual countries.

Brazil is much larger in levels, but it is less dependent on intra-bloc trade shares. In 2017, Brazil exports USD 216.5 billion in the model baseline and sends 10.4 percent of exports to other Mercosur members; Uruguay exports USD 10.4 billion and sends 18.5 percent to the bloc, while 31.7 percent of its imports come from Mercosur. Paraguay is even more exposed to the regional market in shares. The point is not that intra-bloc trade alone explains the bloc's stability. Rather, the baseline structure shows why small members can value membership even when most of their exports go outside the bloc: leaving or being excluded affects a larger share of their observed trade, and membership also changes their external bargaining position.

Table 8: Baseline trade levels, intra-bloc exposure, and export concentration

Panel A. Trade levels and intra-bloc exposure					
Country	Year	Exports	Imports	Intra-bloc X	Intra-bloc M
		USD billions		Percent of total	
Paraguay	2004	2.6	2.8	32.4	45.3
Paraguay	2017	6.9	10.0	31.7	38.3
Uruguay	2004	3.5	3.5	23.4	46.7
Uruguay	2017	10.4	9.0	18.5	31.7
Argentina	2004	30.9	20.6	18.2	38.1
Argentina	2017	59.4	56.3	19.5	34.6
Brazil	2004	97.1	53.0	9.3	10.5
Brazil	2017	216.5	138.2	10.4	8.8
Panel B. Export composition and partner concentration					
Country	Year	Top export sector	Main intra-bloc partner	Main external partner	
Paraguay	2004	Oil seeds (52.3)	Uruguay (16.0)	Rest of world (25.3)	
Paraguay	2017	Oil seeds (34.1)	Brazil (17.7)	Rest of world (20.9)	
Uruguay	2004	Meat: cattle,sheep,goats,horse (18.5)	Brazil (14.9)	European Union (26.2)	
Uruguay	2017	Paper products, publishing (17.2)	Brazil (12.9)	China (27.8)	
Argentina	2004	Vegetable oils and fats (22.1)	Brazil (15.4)	European Union (23.0)	
Argentina	2017	Vegetable oils and fats (26.7)	Brazil (16.1)	Rest of world (31.4)	
Brazil	2004	Motor vehicles and parts (10.7)	Argentina (7.7)	European Union (26.0)	
Brazil	2017	Mineral products nec (18.8)	Argentina (8.4)	China (25.3)	

Notes: Trade levels exclude domestic transactions and are reported in USD billions from the GTAP model baseline. Intra-bloc exports and imports are shares of each country's total exports and imports with foreign partners. Main intra-bloc and external partners report the partner's share of total exports. Sector labels use the paper's 33-sector aggregation. The table reports the first and last benchmark years; the complete five-year series is saved in the replication tables.

A.3 Sectoral Exposure Metrics and Top Drivers

This appendix defines the concentration metrics used in Table 5. Let b_{is} be country i 's baseline labor share in sector s , and let $\Delta\ell_{is}$ be the sectoral labor-share change in percentage points under the reciprocal bloc counterfactual. Define the gain and loss sector sets as

$$S_i^+ = \{s : \Delta\ell_{is} > 0\}, \quad S_i^- = \{s : \Delta\ell_{is} < 0\}.$$

For each side $h \in \{+, -\}$, define the side-specific exposure share

$$q_{is}^h = \frac{|\Delta\ell_{is}|}{\sum_{r \in S_i^h} |\Delta\ell_{ir}|}, \quad s \in S_i^h.$$

The baseline effective number is

$$N_i^{base} = \frac{1}{\sum_s b_{is}^2},$$

and the effective number of exposed sectors is

$$N_{ih}^{exposure} = \frac{1}{\sum_{s \in S_i^h} (q_{is}^h)^2}.$$

The concentration ratio in Table 5 is $N_i^{base} / N_{ih}^{exposure}$. A value above one means that the gain or loss exposure is more concentrated than the country's own baseline labor allocation. The top-sector share is $\max_{s \in S_i^h} q_{is}^h$, and the top-sector baseline share is the corresponding b_{is} for that sector. Total labor-share adjustment is $\sum_{s \in S_i^h} |\Delta\ell_{is}|$ and is reported in percentage points.

Table 9: Top sectoral drivers of approval-margin exposure: China 2017

Country	Side	Rank	Sector	Δ LS	Shock share	Base share	Export share	Import share	Partner cut	Own cut
Brazil	Gain	1	Meat: cattle, sheep, goats, horse	0.13	33.8	4.8	1.6	0.2	12.0	1.7
Brazil	Gain	2	Oil seeds	0.12	31.1	3.7	41.1	0.0	3.0	1.0
Brazil	Gain	3	Plant-based fibers	0.08	19.5	0.7	0.3	0.0	18.9	0.0
Brazil	Gain	4	Meat products nec	0.02	4.8	2.0	2.0	0.0	6.8	2.7
Brazil	Gain	5	Raw and processed sugar	0.01	3.6	1.6	0.7	0.0	50.0	0.0
Brazil	Loss	1	Mineral products nec	-0.12	29.4	9.6	37.7	2.0	0.1	2.7
Brazil	Loss	2	Wearing apparel	-0.07	17.2	2.8	0.0	5.5	16.6	8.1
Brazil	Loss	3	Manufactures nec	-0.05	13.2	6.1	0.1	5.6	8.7	4.4
Brazil	Loss	4	Chemical, rubber, plastic prods	-0.05	12.7	15.9	1.1	14.9	7.8	2.2
Brazil	Loss	5	Electronic equipment	-0.02	5.5	1.4	0.2	23.5	2.9	2.0
Uruguay	Gain	1	Meat: cattle, sheep, goats, horse	2.78	91.5	12.4	25.0	0.0	12.0	2.9
Uruguay	Gain	2	Textiles	0.12	4.0	1.2	2.8	3.5	22.2	4.3
Uruguay	Gain	3	Wool, silk-worm cocoons	0.08	2.5	0.2	1.0	0.0	38.0	0.0
Uruguay	Gain	4	Oil seeds	0.05	1.7	4.3	38.5	0.0	3.0	1.6
Uruguay	Gain	5	Animal products nec	0.01	0.3	0.5	1.7	0.0	6.3	2.9
Uruguay	Loss	1	Chemical, rubber, plastic prods	-0.57	18.9	8.3	0.0	16.7	6.8	2.5
Uruguay	Loss	2	Manufactures nec	-0.39	12.8	9.9	0.1	10.3	15.7	4.2
Uruguay	Loss	3	Cattle, sheep, goats, horses	-0.33	10.9	9.5	0.4	0.0	9.8	0.0
Uruguay	Loss	4	Leather products	-0.26	8.5	2.1	1.0	4.0	6.3	6.9
Uruguay	Loss	5	Mineral products nec	-0.24	8.0	3.6	0.2	1.9	3.1	3.1

Notes: Δ LS is the sectoral labor-share change in percentage points. Exposure share is the sector's share of total absolute movement on the gain or loss side. Base share is the sector's baseline labor share. Export share is the sector's share of the country's exports to the partner. Import share is the sector's share of imports from the partner. Partner cut is the partner's tariff cut faced by the country's exports. Own cut is the Mercosur member's tariff cut on imports from the partner. Shares are percentages; tariff cuts are percentage points.

Table 10: Top sectoral drivers of approval-margin exposure: United States 2004

Country	Side	Rank	Sector	Δ LS	Shock share	Base share	Export share	Import share	Partner cut	Own cut
Brazil	Gain	1	Leather products	0.41	47.9	2.5	6.8	0.1	7.2	17.8
Brazil	Gain	2	Mineral products nec	0.15	17.8	3.1	5.6	0.7	3.3	10.7
Brazil	Gain	3	Crops nec	0.09	11.1	2.6	2.8	0.1	29.4	5.8
Brazil	Gain	4	Wood products	0.05	5.5	2.5	11.0	0.2	0.6	16.7
Brazil	Gain	5	Metals nec	0.04	4.4	2.1	6.9	0.9	0.4	9.1
Brazil	Loss	1	Chemical, rubber, plastic prods	-0.51	59.8	18.4	6.2	31.9	1.4	9.7
Brazil	Loss	2	Electronic equipment	-0.15	17.3	6.4	2.7	17.2	0.4	9.8
Brazil	Loss	3	Machinery and equipment nec	-0.05	6.0	4.9	10.8	21.0	0.3	12.5
Brazil	Loss	4	Manufactures nec	-0.04	4.5	2.8	0.8	0.6	0.0	18.6
Brazil	Loss	5	Motor vehicles and parts	-0.04	4.3	8.2	7.8	4.5	1.2	15.6
Uruguay	Gain	1	Meat: cattle, sheep, goats, horse	5.67	78.5	12.9	63.2	0.1	15.3	12.5
Uruguay	Gain	2	Raw and processed dairy	1.55	21.5	6.8	3.6	0.2	52.7	17.0
Uruguay	Loss	1	Chemical, rubber, plastic prods	-1.91	26.4	10.9	1.6	31.4	0.4	9.8
Uruguay	Loss	2	Leather products	-1.40	19.4	6.7	7.5	0.5	0.5	9.2
Uruguay	Loss	3	Mineral products nec	-0.40	5.6	2.3	1.0	0.4	1.8	12.3
Uruguay	Loss	4	Manufactures nec	-0.36	5.0	2.5	0.9	3.5	0.0	20.3
Uruguay	Loss	5	Wood products	-0.35	4.8	2.4	5.2	0.3	0.1	16.0

Notes: Δ LS is the sectoral labor-share change in percentage points. Exposure share is the sector's share of total absolute movement on the gain or loss side. Base share is the sector's baseline labor share. Export share is the sector's share of the country's exports to the partner. Import share is the sector's share of imports from the partner. Partner cut is the partner's tariff cut faced by the country's exports. Own cut is the Mercosur member's tariff cut on imports from the partner. Shares are percentages; tariff cuts are percentage points.

A.4 Argentina and Paraguay Sectoral Concentration Diagnostics

The main text uses Brazil and Uruguay as focal cases, while this appendix reports Argentina and Paraguay to preserve the all-member approval record. The same baseline-adjusted measure is still part of the empirical accounting. For Argentina, the aggregate reciprocal welfare effects are small or negative in the focal cases, and the gain- and loss-side concentration patterns reported here are the sectoral inputs behind the approval-margin interpretation.

Paraguay's results vary across the two focal cases. In the United States 2004 case, it gains 0.490 percent under the bloc reciprocal counterfactual, and gain-side exposure is highly concentrated in raw and processed sugar. In the China 2017 case, its reciprocal bloc effect is close to zero under the main calibration, at 0.001 percent, with gains and losses spread across different agricultural and manufacturing sectors. The sign of the China result also changes across the alternative elasticity calibrations reported in Appendix A.7. These results warrant caution in interpreting Paraguay's incentives. Geography and transit costs are outside the model, so the exercise cannot identify their contribution to the value of membership.

Table 11: Appendix-member approval-margin sectoral exposure under reciprocal tariff cuts

Scenario	Country	Side	Welfare	Realloc.	Base EffN	Shock EffN	Ratio	Top sector	Top share	Base share
China 2017	Argentina	Gain	0.094	0.34	18.89	1.91	9.87	Meat: cattle, sheep, goats, horse	69.5	5.5
China 2017	Argentina	Loss	0.094	0.34	18.89	9.83	1.92	Manufactures nec	18.6	2.6
China 2017	Paraguay	Gain	0.001	0.13	17.50	5.51	3.18	Meat: cattle, sheep, goats, horse	34.0	6.1
China 2017	Paraguay	Loss	0.001	0.13	17.50	3.49	5.02	Chemical, rubber, plastic prods	47.5	15.0
United States 2004	Argentina	Gain	-0.077	0.63	17.60	5.39	3.26	Raw and processed dairy	37.4	4.8
United States 2004	Argentina	Loss	-0.077	0.63	17.60	1.95	9.03	Chemical, rubber, plastic prods	70.0	15.3
United States 2004	Paraguay	Gain	0.490	0.86	12.45	2.81	4.43	Raw and processed sugar	53.9	1.0
United States 2004	Paraguay	Loss	0.490	0.86	12.45	3.59	3.47	Manufactures nec	49.4	2.2

Notes: Welfare and exposure both use the WITS-main calibration. Welfare is the reciprocal bloc real-income change in percent. Realloc. is the total absolute labor-share movement on the gain or loss side, in percentage points. Base EffN and Exposure EffN are inverse-Herfindahl effective numbers of sectors for the baseline labor allocation and the exposure-side distribution. Ratio is Base EffN divided by Exposure EffN. Top share and Base share are percentages. Appendix A.3 gives formulas. Argentina and Paraguay are reported here to complete the all-member comparison. Paraguay's outcomes vary across scenarios, and its China result is sensitive to the elasticity calibration; the model does not identify geography or transit-cost effects.

A.5 Institutional Timeline

Exit and discipline operate through a related but distinct institutional margin. Accession is open to ALADI members but requires unanimity, domestic ratification, adoption of the common external tariff, and incorporation of Mercosur obligations. Venezuela entered through this route and is now suspended; Bolivia ratified in 2024 and is in transition (MERCOSUR, 2024, 2026). Suspension has been used for democratic or accession-compliance disputes, as in Paraguay in 2012 and Venezuela in 2016–2017, rather than as an automatic sanction for ordinary trade-policy disagreement (MERCOSUR Common Market Council, 2012; Ministry of Foreign Affairs of Argentina, 2016; MERCOSUR, 2017). The treaties regulate voluntary withdrawal, and the Protocol of Ushuaia allows suspension when democratic order breaks down (MERCOSUR, 1998). Institutional status rules govern exit and discipline, while collective negotiating authority and consensus shape approval of new external agreements.

A.5.1 Timeline mapping for Mercosur policy debates

Table 12: Timeline evidence: negotiations, agreements, and internal stress

Timeline track	Main episodes	Interpretation for the paper
EU–Mercosur negotiations	Negotiations launched in 1999; agreement concluded in principle in 2019; political agreement announced in 2024; the EMPA and interim trade agreement were signed in January 2026; the interim trade agreement has provisionally applied since May 2026, while full EMPA entry into force still depends on full ratification.	Maps to the external gridlock mechanism. The bloc can create bargaining power in large negotiations, but agreement progress does not remove domestic and internal approval frictions.
Other external agreements	Mercosur–Singapore was signed in 2023; talks with Canada and Korea remained on the agenda; Mercosur–UAE negotiations were launched in 2024; EFTA negotiations were concluded and signed in 2025.	Maps to bloc bargaining capacity. Mercosur is not generally unable to negotiate; gridlock is sharper when liberalization is large, politically sensitive, or distributionally uneven.
Threats and internal stress	Uruguay explored a United States track in 2004–2006 and later signaled an independent China path in 2021–2022; Venezuela was suspended in 2016–2017; Bolivia advanced accession in 2024; later unilateral-option debates in Argentina, Brazil, and Uruguay kept the same legal tension visible.	Maps to internal stability and exit pressure. Exit threats and unilateral pressure can be politically salient even when the bloc remains institutionally stable.

Notes. The table summarizes the timeline slide from the Midwest Trade Conference presentation and the interactive presentation at <https://presentations-pied.vercel.app/>. Dates summarize the presentation material and are used here to map policy episodes into the paper’s mechanisms rather than to provide a full diplomatic history.

A.6 Policy Notes and Institutional Sources

A.6.1 Institutional source tables

Table 13: Core legal instruments for Mercosur's institutional mechanics

Instrument	Date and place	Why it matters	Main source
Treaty of Asuncion	March 26, 1991, Asuncion	Founding treaty. Article 1 sets the common-market program, including the common external tariff and common commercial policy. Articles 20–22 regulate accession and denunciation.	(MERCOSUR, 1991)
Protocol of Ouro Preto	December 17, 1994, Ouro Preto	Gives Mercosur international legal personality and defines the institutional structure, consensus rule, domestic incorporation procedure, and binding effect of norms.	(MERCOSUR, 1994)
Protocol of Ushuaia	July 24, 1998, Ushuaia	Democratic clause. It provides the legal basis for consultations and suspension when democratic order breaks down.	(MERCOSUR, 1998)
Protocol of Olivos	February 18, 2002, Olivos	Current dispute-settlement framework for controversies over the Treaty of Asuncion, Mercosur norms, and agreements concluded within the bloc.	(MERCOSUR, 2002)
CMC Decision 32/00	June 29, 2000, Buenos Aires	Reaffirms joint negotiation of tariff-preference agreements; read alongside the incorporation-status qualification in Section 2.	(MERCOSUR Common Market Council, 2000)
CMC Decisions 28/05 and 20/19	December 8, 2005, Montevideo; December 4, 2019, Bento Goncalves	Operational accession rules. Decision 20/19 links voice and voting rights to reciprocity, compliance, and gradual incorporation of the Mercosur acquis.	(MERCOSUR Common Market Council, 2005, 2019)

Table 14: Selected institutional episodes for the negotiation narrative

Episode	Date	Institutional relevance	Main source
Paraguay suspension	2012	Application of the democratic-clause framework after the removal of President Fernando Lugo. The episode shows that suspension exists for specific political conditions, not as a routine response to trade-policy disagreement.	(MERCOSUR Common Market Council, 2012)
Venezuela suspension	2016–2017	Venezuela first lost the exercise of member rights after noncompliance with accession obligations and was later suspended under the Protocol of Ushuaia.	(MERCOSUR, 2017)
Bolivia accession	2015–2024	Bolivia's accession illustrates gradual membership: the protocol was signed in 2015, ratification was completed in 2024, and full participation depends on incorporation of the acquis.	(MERCOSUR, 2024)
Uruguay outside-option debates	2006–2007; 2021–2026	Uruguay's United States and China initiatives motivate comparing national and bloc routes to external market access.	(Office of the United States Trade Representative, 2007; Presidency of Uruguay, 2022; Governments of Uruguay and China, 2026)
EU–Mercosur agreement	1999–2026	The largest external negotiation shows the bargaining value of acting as a bloc, but also the role of ratification, provisional application, and domestic approval frictions after signature.	(European Commission, 2026)

A.6.2 Negotiation documents and the China chronology

The Mercosur–India agreement and the Mercosur–Japan framework support different parts of the negotiating interpretation. Article 3 of the India agreement specifies concessions from each side; Article 35 left its annexes to

subsequent negotiation, so the 2004 text should not be treated as a completed tariff schedule. The 2026 Japan framework states a balanced-package objective. Using these precedents for the China exercise is an analogy about exchanging market access, rather than a China-specific mandate or a numerical reciprocity requirement ([Mercosur and India, 2004](#); [Mercosur, 2026b](#)). WTO compatibility is a separate question from the tariff-selection rule used here.

Table 15: Uruguay–China initiatives relevant to the negotiating alternatives

Date	Documented step	Implication for the exercise
September 2021	Uruguay organizes the joint feasibility-study team (Ministry of Foreign Affairs of Uruguay, 2021).	Preparatory work for a possible agreement.
July 2022	Uruguay announces intended formal dialogue; China confirms completion of the study and further communication (Presidency of Uruguay, 2022 ; Ministry of Commerce of China, 2022).	Distinguishes an announcement from evidence of exchanged tariff offers.
November 2023	Joint declaration addresses free-trade cooperation, Mercosur dialogue and an investment-treaty update (Ministry of Foreign Affairs of Uruguay, 2023).	Bilateral cooperation and a possible bloc route coexist.
February 2026	Joint declaration expresses an aspiration to begin Mercosur–China FTA talks alongside bilateral investment work (Governments of Uruguay and China, 2026).	Proposed trade talks and investment negotiations concern different instruments.

The reviewed documents do not supply an agreed tariff schedule matching the model or identify a single cause of delay. Dates describe diplomatic initiatives, not the years of the model’s benchmark economies.

A.6.3 CET departures: coverage and descriptive trade comparisons

The country collection distinguishes legal tariff schedules, complete recovered product baskets and customs observations. An NCM8 code is an eight-digit product category; eligibility can be narrower because of product descriptions, end-use conditions or quotas. A same-date gap between the CET and a national rate measures a departure from the reference schedule. A policy cut instead compares national rates before and after a change for the same eligible scope. Missing prior rates prevent treating every newly documented zero rate as a measured cut of known size.

Table 16: Recovered national exceptions and special tariff arrangements

Country / date	National list or full schedule	Other recovered below-CET entries
Argentina October 2023	National list: 92 of 102 rate entries below the CET; the list covers 100 distinct codes.	Capital goods: 344 of 374 entries. IT: 144 of 146. Separate annexes also contain higher rates (Government of Argentina, 2023).
Brazil September 2026	National list: 320 of 352 scope/rate entries below the CET. Codes can recur for different descriptions or quota periods.	Shortage relief: 146 of 146 entries. The capital/IT list is not a census of product-specific concessions (Ministry of Development, Industry, Trade and Services of Brazil, 2026).
Paraguay October 2025	National-list markers: 690 of 722 entries below the CET.	Capital goods: 943 of 1,250; IT: 220 of 361; textiles: 807 of 827. Regime markers overlap (Ministry of Economy and Finance of Paraguay, 2025).
Uruguay February 2025	Full national schedule: 4,204 of 12,243 entries below the CET (34.34%), across multiple regimes.	Six recovered 2017 lower-rate lists cover 2,265 distinct eight-digit product codes. That earlier union of lists is a different snapshot (Ministry of Economy and Finance of Uruguay, 2025 ; Government of Uruguay, 2016b).

Source: author's country-report extractions from the cited official schedules. An entry is a published row or a row carrying a regime marker; it is not necessarily a distinct product. A code can have more than one eligible scope. Counts describe documented treatment at different dates, not imports actually using an exemption or newly granted tariff cuts. Regimes overlap, so rows cannot be added or used to rank countries' economic reliance on exemptions. Broad schedule snapshots have not been independently purged of pandemic-origin provisions; COVID-response events remain excluded from interpreted before-and-after comparisons.

The extraction preserves the distinction between what a measure changes and how many entries appear in a later schedule. Uruguay's Decree 237/2017 adds five zero-rate entries, amends the scope of an existing exception and removes five entries ([Government of Uruguay, 2017](#), Articles 1–3). The six-code lower-rate basket analyzed below includes the additions and scope amendment, not the removals. Similarly, Brazil's product-specific capital and IT concessions can cover only a description within a tariff code and depend on the absence of equivalent domestic production ([Ministry of Development, Industry, Trade and Services of Brazil, 2024](#)). Such concessions are not fully counted by the recovered Brazilian capital/IT list. The inventories are documented legal treatment, rather than a complete census of shipment-level use.

Table 17: Uruguay: common zero rates, remaining protection and national departures

Treatment in the February 2025 schedule	Entries	Share (%)
National rate below the CET, but still positive	3,205	26.18
National rate below the CET and equal to zero	999	8.16
National rate equals a positive CET	4,961	40.52
National rate and CET both equal zero	3,008	24.57
National rate above the CET	69	0.56
Unresolved rate	1	0.01
Total published entries	12,243	100.00

Source: author's Uruguay report, calculated from the February 2025 national schedule ([Ministry of Economy and Finance of Uruguay, 2025](#)). The first two rows sum to 4,204 entries below the CET (34.34%). Shares use all 12,243 published entries, including the unresolved entry. They are not import-value shares. The classification does not identify why each rate is zero or establish production conditions for each good.

The table separates the presence of positive protection from compliance with a common rate. A common zero tariff gives no scope for an additional tariff cut, while a positive national rate below the CET still leaves protection to negotiate. Counting either as simply outside an “effective CET” would obscure that difference. Economic relevance also depends on import values, the size of the rate difference and competing production within Mercosur.

Current import weights alone can understate the relevance of highly protected goods when protection suppresses imports. The recovered schedules do not establish that every zero rate reflects a production gap, or estimate how much regional trade or production depends on coordinated tariffs.

The legal search covers 2006–2026 but does not reconstruct a complete product-by-date history for those twenty years. The trade comparisons retain all codes in each recovered measure, including those with no recorded imports. Table 18 lists all eight measured non-COVID baskets. Argentina’s Decree 117/2017 has two annexes with different implementation dates. Brazil’s 709/2025 package includes nine additions, a sardine quota and a palm-oil quota expansion; palm oil already had a zero in-quota rate ([Government of Argentina, 2017](#); [Ministry of Development, Industry, Trade and Services of Brazil, 2025](#)). The five Uruguay measures include additions and changes in scope ([Government of Uruguay, 2016a, 2017, 2018, 2019, 2021](#)). These differences preclude describing every entry as a new uniform tariff cut.

Table 18: Complete recovered non-COVID baskets: import-value comparisons

Measure	Codes	Before	After	US\$ million		Change (%)
				Before	After	
Argentina: 117/2017, Annex I	60	2016-11 2017-01	2017-03 2017-05	160.865	191.789	+19.2
Argentina: 117/2017, Annex II	12	2016-12 2017-02	2017-04 2017-06	69.606	170.734	+145.3
Brazil: 709/2025	11	2024-12 2025-02	2025-04 2025-06	339.113	369.664	+9.0
Uruguay: 242/2016	1	2016-08 2016-09	2016-11 2016-12	0.008	0.008	-3.1
Uruguay: 237/2017	6	2017-06 2017-08	2017-10 2017-12	9.174	11.615	+26.6
Uruguay: 129/2018	2	2018-03 2018-05	2018-07 2018-09	4.672	4.084	-12.6
Uruguay: 143/2019	1	2019-04 2019-06	2019-08 2019-10	0.189	0.205	+8.4
Uruguay: 44/2021	1	2020-12 2021-02	2021-04 2021-06	0.386	0.184	-52.5

Dates give the first and last complete months in each window; the event month is omitted. Uruguay 242/2016 uses two months per side, the other measures three. Values are current-dollar CIF for Argentina and Brazil, and customs assessment values from initial filings for Uruguay. The measures and all listed NCM8 codes are fixed before comparing outcomes. Sources: INDEC, MDIC Comex Stat and Uruguay customs; legal instruments are cited in the text. Paraguay has no completed non-COVID comparison in the recovered data.

Exclusion follows the measure’s documented purpose: COVID-response sources remain in the data but do not enter these comparisons. Uruguay’s ordinary 44/2021 microfiber amendment remains included, with its pandemic-period window stated explicitly. The pre-2020 subset consists of the two Argentine measures and Uruguay’s 2016–2019 measures; it already includes both rising and declining import values and different leading supplier gains. The later Brazilian and Uruguayan cases are not needed to establish that the observed patterns differ. Paraguay’s missing non-COVID comparison is a coverage gap, rather than evidence of no trade response.

The short windows combine tariff changes with seasonality, exchange rates, commodity prices, demand and supply conditions. The alternative six-month windows change the leading share gainer from Thailand to Taiwan for Argentina’s Annex I and from Vietnam to Portugal for Brazil’s food package. Uruguay’s 2016 basket begins from only US\$8,390, and the 2018 basket has mixed May/June implementation timing. The comparisons therefore motivate further investigation without identifying trade creation, trade diversion or welfare effects. A lower external tariff can reduce an existing preference wedge; whether it increases or reduces an economic distortion depends on the stated counterfactual.

Recorded duty ratios provide a separate check on the timing of national measures (Table 19). The large fall for Uruguay's 2017 basket occurs alongside supplier changes, whereas the 2018 basket does not show a corresponding fall. These are averages across recorded customs treatment, not a classification of every shipment's legal regime (National Customs Directorate of Uruguay, 2026). A further 2017 diagnostic finds zero recorded duties against a positive base-CET reference for 34.88% of external import value, and positive recorded duties materially below that reference for 17.13%, using a US\$5.927 billion denominator. Preferences, special regimes, reference dates and initial-filing amounts prevent attributing these shares entirely to national exemptions or treating them as final net payments. For the 2017 basket, the duty/value ratio falls from 10.93 to 1.71 percent between June–August and October–December 2017, using 418 and 506 usable external-origin records respectively. This diagnostic is descriptive; it does not establish the causal effect of the decree or certified final payments.

Table 19: Uruguay: recorded duty ratios around the retained measures

Decree	Duty/value (%)		Records	Scope
	Before	After		
242/2016	0.00	0.00	1/2	One zero-rate addition; very few usable records.
237/2017	10.93	1.71	418/506	Five additions and one scope expansion; eligibility remains conditional.
129/2018	0.45	0.50	38/38	Addition and scope revision; mixed May/June implementation.
143/2019	14.00	0.00	10/8	One zero-rate addition; small observation counts.
44/2021	1.95	0.00	33/6	Microfiber subset; prior zero-duty imports and a code split.

Numerators are recorded duties in initial customs filings; denominators are the value of identified external-origin item records with usable duty data. Other preferences are included. These basket ratios are neither statutory rates nor verified final net payments. Windows are those in Table 18; they do not identify individual exemption use. The public Argentina and Brazil trade extracts do not provide comparable duty observations.

Supplier accounting for each complete basket

The following tables show the six largest absolute supplier-share changes where at least six origins are observed. Every remaining origin stays in the residual and in the denominator. Dollar values are reported alongside shares because a supplier can lose share while selling more. Qualifying regional goods generally already receive preferential treatment, so changes in Brazilian, Argentine or Paraguayan sales do not establish that those goods received a new CET cut. Rules of origin and special arrangements still determine eligibility.

Table 20: Argentina: 117/2017, Annex I: supplier shares and import values

Supplier	US\$ thousands		Share (%)		Change (pp)
	Before	After	Before	After	
China	93,755.847	101,099.048	58.28	52.71	-5.57
Thailand	8,335.186	14,423.984	5.18	7.52	+2.34
Israel	4,462.280	889.067	2.77	0.46	-2.31
Taiwan	6,520.719	12,198.637	4.05	6.36	+2.31
Hungary	1,053.286	3,667.609	0.65	1.91	+1.26
Philippines	3,170.092	5,892.943	1.97	3.07	+1.10
Other origins	43,567.776	53,618.212	27.08	27.96	+0.87
All origins	160,865.187	191,789.500	100.00	100.00	0.00

Complete basket: 60 NCM8 codes. Before: 2016-11 to 2017-01; after: 2017-03 to 2017-05. Amounts are US\$ thousands; shares use all basket imports. Up to six origins with the largest absolute share changes are shown separately; Other origins retains the remainder. Values are rounded only for display. These are descriptive changes, not estimated tariff effects.

Table 21: Argentina: 117/2017, Annex II: supplier shares and import values

Supplier	US\$ thousands		Share (%)		Change (pp)
	Before	After	Before	After	
China	37,514.906	130,846.442	53.90	76.64	+22.74
Mexico	16,631.747	22,660.707	23.89	13.27	-10.62
United States	8,165.737	6,901.856	11.73	4.04	-7.69
Brazil	2,361.304	1,355.737	3.39	0.79	-2.60
Germany	1,253.919	812.713	1.80	0.48	-1.33
Japan	364.773	187.889	0.52	0.11	-0.41
Other origins	3,313.658	7,968.395	4.76	4.67	-0.09
All origins	69,606.046	170,733.738	100.00	100.00	0.00

Complete basket: 12 NCM8 codes. Before: 2016-12 to 2017-02; after: 2017-04 to 2017-06. Amounts are US\$ thousands; shares use all basket imports. Up to six origins with the largest absolute share changes are shown separately; Other origins retains the remainder. Values are rounded only for display. These are descriptive changes, not estimated tariff effects.

Table 22: Brazil: 709/2025: supplier shares and import values

Supplier	US\$ thousands		Share (%)		Change (pp)
	Before	After	Before	After	
Paraguay	82,414.684	58,672.229	24.30	15.87	-8.43
Vietnam	50.098	26,978.767	0.01	7.30	+7.28
Spain	10,576.936	21,856.876	3.12	5.91	+2.79
Chile	7,780.075	1,822.649	2.29	0.49	-1.80
Indonesia	72,746.658	84,402.365	21.45	22.83	+1.38
Australia	8,227.559	3,957.405	2.43	1.07	-1.36
Other origins	157,317.172	171,973.723	46.39	46.52	+0.13
All origins	339,113.182	369,664.014	100.00	100.00	0.00

Complete basket: 11 NCM8 codes. Before: 2024-12 to 2025-02; after: 2025-04 to 2025-06. Amounts are US\$ thousands; shares use all basket imports. Up to six origins with the largest absolute share changes are shown separately; Other origins retains the remainder. Values are rounded only for display. These are descriptive changes, not estimated tariff effects.

Table 23: Uruguay: 242/2016: supplier shares and import values

Supplier	US\$ thousands		Share (%)		Change (pp)
	Before	After	Before	After	
China	0.000	7.532	0.00	92.69	+92.69
Peru	7.444	0.594	88.72	7.31	-81.41
Argentina	0.946	0.000	11.28	0.00	-11.28
All origins	8.390	8.127	100.00	100.00	0.00

Complete basket: 1 NCM8 codes. Before: 2016-08 to 2016-09; after: 2016-11 to 2016-12. Amounts are US\$ thousands; shares use all basket imports. Up to six origins with the largest absolute share changes are shown separately; Other origins retains the remainder. Values are rounded only for display. These are descriptive changes, not estimated tariff effects.

Table 24: Uruguay: 237/2017: supplier shares and import values

Supplier	US\$ thousands		Share (%)		Change (pp)
	Before	After	Before	After	
Spain	185.298	1,849.641	2.02	15.92	+13.90
China	2,684.365	2,209.472	29.26	19.02	-10.24
Brazil	3,262.403	3,316.899	35.56	28.56	-7.01
Israel	198.865	846.317	2.17	7.29	+5.12
France	17.769	194.104	0.19	1.67	+1.48
Argentina	1,856.138	2,196.354	20.23	18.91	-1.32
Other origins	968.933	1,002.622	10.56	8.63	-1.93
All origins	9,173.770	11,615.410	100.00	100.00	0.00

Complete basket: 6 NCM8 codes. Before: 2017-06 to 2017-08; after: 2017-10 to 2017-12. Amounts are US\$ thousands; shares use all basket imports. Up to six origins with the largest absolute share changes are shown separately; Other origins retains the remainder. Values are rounded only for display. These are descriptive changes, not estimated tariff effects.

Table 25: Uruguay: 129/2018: supplier shares and import values

Supplier	US\$ thousands		Share (%)		Change (pp)
	Before	After	Before	After	
Brazil	1,688.569	1,257.282	36.14	30.79	-5.36
United States	1,267.416	1,281.448	27.13	31.38	+4.25
South Korea	0.000	163.671	0.00	4.01	+4.01
Argentina	863.054	619.265	18.47	15.16	-3.31
Singapore	119.388	0.000	2.56	0.00	-2.56
Chile	31.693	117.149	0.68	2.87	+2.19
Other origins	701.649	645.093	15.02	15.80	+0.78
All origins	4,671.769	4,083.907	100.00	100.00	0.00

Complete basket: 2 NCM8 codes. Before: 2018-03 to 2018-05; after: 2018-07 to 2018-09. Amounts are US\$ thousands; shares use all basket imports. Up to six origins with the largest absolute share changes are shown separately; Other origins retains the remainder. Values are rounded only for display. These are descriptive changes, not estimated tariff effects.

Table 26: Uruguay: 143/2019: supplier shares and import values

Supplier	US\$ thousands		Share (%)		Change (pp)
	Before	After	Before	After	
South Korea	106.721	0.000	56.41	0.00	-56.41
Spain	58.777	124.175	31.07	60.56	+29.49
United States	0.000	37.917	0.00	18.49	+18.49
Brazil	0.036	29.019	0.02	14.15	+14.13
Germany	22.081	13.948	11.67	6.80	-4.87
China	1.574	0.000	0.83	0.00	-0.83
All origins	189.189	205.058	100.00	100.00	0.00

Complete basket: 1 NCM8 codes. Before: 2019-04 to 2019-06; after: 2019-08 to 2019-10. Amounts are US\$ thousands; shares use all basket imports. Up to six origins with the largest absolute share changes are shown separately; Other origins retains the remainder. Values are rounded only for display. These are descriptive changes, not estimated tariff effects.

Table 27: Uruguay: 44/2021: supplier shares and import values

Supplier	US\$ thousands		Share (%)		Change (pp)
	Before	After	Before	After	
Spain	1.329	0.000	0.34	0.00	-0.34
China	384.966	183.622	99.66	100.00	+0.34
All origins	386.295	183.622	100.00	100.00	0.00

Complete basket: 1 NCM8 codes. Before: 2020-12 to 2021-02; after: 2021-04 to 2021-06. Amounts are US\$ thousands; shares use all basket imports. Up to six origins with the largest absolute share changes are shown separately; Other origins retains the remainder. Values are rounded only for display. These are descriptive changes, not estimated tariff effects.

A.6.4 Why the two margins differ

The model separates internal stability from external gridlock. A simple numerical illustration makes the distinction clear:

- If a member obtains $G_i(0) = 1.20$ inside the customs union and $G_i(out) = 1.05$ under the outside option, the member prefers staying to this outside option; union-wide stability requires the corresponding condition for every member.
- If the same member obtains $G_i(agree) = 1.12$ under the external agreement while the status quo gives $G_i(0) = 1.20$, then the all-member approval condition fails even though the member remains inside the union.
- If the agreement instead raises the payoff to $G_i(agree) = 1.25$, the member has a positive approval margin and supports liberalization.

These examples show why the paper needs both margins: participation stability and negotiation feasibility are not the same margin.

A.7 Elasticity Calibration

The main quantitative exercises use a single sectoral elasticity vector estimated from WITS/TRAINS-BACI tariff and trade data mapped into the paper's 33-sector model aggregation. This appendix records the calculation, the data caveats, and the comparison with the Ossa benchmark vector and the direct GTAP check.

Data choice. The model is solved on the GTAP/Ossa database because that database gives a balanced applied-general-equilibrium environment: sectoral trade flows, applied protection, expenditure shares, and production accounts are internally consistent. That same feature makes GTAP less attractive for estimating elasticities. In the 16-region $\times$ 33-sector aggregation used here, product-line and country-pair tariff variation is already compressed. The pooled GTAP triple-ratio regression therefore identifies only a small number of sectors directly.

WITS/TRAINS is more useful for the elasticity step because it contains detailed tariff-line information before the data are mapped into the 33-sector model grid. The WITS data include applied MFN tariffs and, when available, preferential tariff regimes. WITS defines effectively applied tariffs as the lowest available tariff when a preference exists, and the MFN applied tariff otherwise. This gives more product and reporter-partner variation before aggregation and reduces the risk that the elasticity vector is driven by the GTAP aggregation itself.

Construction. I start from the WITS/TRAINS-BACI tariff panel shared with the EU–Mercosur project and map HS6 products into the 33-sector model aggregation using transparent HS2/HS4 concordance rules. Within each year-sector-reporter-exporter cell, HS6 tariffs are aggregated using trade values as weights. The trade-flow

dependent variable uses BACI values without tariff payments: let $x_{ab,s}$ denote producer-valued sales from exporter a to importer b . I then estimate the triple-ratio equation sector by sector under this flow-value convention:

$$\log \frac{x_{ab,s}x_{bc,s}x_{ca,s}}{x_{ac,s}x_{cb,s}x_{ba,s}} = -\sigma_s \log \frac{(1+t_{ab,s})(1+t_{bc,s})(1+t_{ca,s})}{(1+t_{ac,s})(1+t_{cb,s})(1+t_{ba,s})} + u_{abc,s}. \quad (12)$$

The estimation files label the negative tariff coefficient as $\theta_s = -\beta_s$. For the producer-valued flows used here, this stored quantity is the CES substitution elasticity σ_s , rather than $\sigma_s - 1$. For tariff-inclusive expenditure, the corresponding coefficient would instead be $1 - \sigma_s$. This distinction reconciles the estimation with the expenditure-share equation in the main text without changing the numerical calibration. The main vector pools the five benchmark years used in the paper, 2004, 2007, 2011, 2014, and 2017, and estimates one elasticity for each sector. A WITS direct estimate enters the main vector only if it is finite, above one, below 100, supported by at least 250 country-triple observations, and has an absolute t -statistic of at least 1.5. If a sector fails this guardrail, I retain the Ossa benchmark value stored in the current SIGMA.csv vector, which already records σ_s and receives no additional conversion. This produces a single WITS-based vector for the baseline calibration while keeping the Ossa vector as the fallback in weakly identified sectors.

Caveats. The WITS data are not a perfect measure of the tariff charged on every shipment. Preferential tariff coverage is incomplete in some years; when preferences are missing, WITS may use MFN rates. WITS also aggregates national tariff lines to HS6 and converts non-ad-valorem tariffs into ad-valorem equivalents where possible. The 2004 panel required the shared WITS/TRAINS-BACI cleaning workflow and includes exact, nearest-prior, preferential, MFN-fallback, and imputation flags. Finally, the HS6-to-33-sector concordance used here is transparent and reproducible but still an aggregation. These caveats are why I keep the Ossa benchmark and the year-specific WITS estimates as robustness checks rather than treating the WITS vector as mechanically definitive.

Why WITS is preferred here. For this paper, WITS is the preferred elasticity source because it identifies many more sectors before fallback, is easier to audit product by product, and uses tariff variation closer to the legal tariff-line data. The direct GTAP exercise is retained as a stored-vector check, but it is too aggregated to be the main calibration source. The result comparisons below show that Argentina, Brazil, and Uruguay retain the signs of reciprocal bloc welfare, own-member welfare, and their difference across the four calibrations in China 2017 and United States 2004, although magnitudes vary. Paraguay retains these signs in United States 2004 but not in China 2017. These comparisons cover the two focal cases, not every partner-year under each alternative vector.

Source documentation. The data interpretation follows the GTAP documentation on effective applied tariff protection by MAcMap and the WITS tariff documentation. The relevant documentation pages are GTAP Data Base documentation (https://www.gtap.agecon.purdue.edu/databases/v12/v12_doco.aspx), GTAP effective applied tariff protection by MAcMap (https://www.gtap.agecon.purdue.edu/resources/res_display.asp?RecordID=7114), WITS About (https://wits.worldbank.org/about_wits.html), WITS bilateral applied tariff technical note (<https://wits.worldbank.org/Bilateral-Tariff-Technical-Note.html>), WITS tariff definitions (https://wits.worldbank.org/WITS/wits/WITSHELP/Content/Data_Retrieval/P/Intro/C2.Types_of_Tariffs.htm), and ITC Market Access Map methodology (<https://m.macmap.org/en/about/methodology>).

Table 28: Elasticity calibration objects

Calibration	Data object	Direct sectors	Fallback sectors	Mean	Median	Min	Max
WITS all-year main	HS6 WITS/TRAINS-BACI mapped to MS33	19	14	5.27	3.29	1.27	13.14
Ossa benchmark	Current SIGMA vector			3.44	2.89	2.12	12.37
GTAP all-year check	GTAP/Ossa 16-region x 33-sector model data	4	29	3.53	2.89	1.04	12.37
WITS early robustness	WITS/MS33 pooled 2004 and 2007	20	13	4.76	3.29	1.46	12.37

Note: Each row is one calibration object. Direct sectors count rows where the sector estimate passes the guardrail and enters directly. Fallback sectors count rows where the vector keeps the Ossa benchmark. The Ossa row is a benchmark vector, not a new regression.

Table 29: Sector elasticities used in the main WITS calibration

ID	Sector	WITS used	WITS direct	Ossa benchmark	Obs	t-stat	Source
1	Raw and processed rice	4.87	0.45	4.87	1609	0.40	Ossa fallback
2	Wheat	12.37	0.46	12.37	264	0.21	Ossa fallback
3	Cereal grains nec	3.29	-0.69	3.29	5250	-2.29	Ossa fallback
4	Vegetables, fruit, nuts	2.19	0.35	2.19	28121	3.26	Ossa fallback
5	Oil seeds	2.89	-0.35	2.89	25286	-3.02	Ossa fallback
6	Raw and processed sugar	1.55	1.55	2.52	25922	10.91	WITS direct
7	Plant-based fibers	10.17	10.17	2.33	4943	10.61	WITS direct
8	Crops nec	2.54	0.20	2.54	35154	1.85	Ossa fallback
9	Cattle,sheep,goats,horses	10.43	10.43	2.58	1482	5.26	WITS direct
10	Animal products nec	2.12	0.69	2.12	30748	3.15	Ossa fallback
11	Raw and processed dairy	5.60	-0.67	5.60	13782	-4.26	Ossa fallback
12	Wool, silk-worm cocoons	2.40	2.40	2.89	17025	5.89	WITS direct
13	Forestry	7.94	7.94	2.33	7808	5.51	WITS direct
14	Meat: cattle,sheep,goats,horse	7.88	7.88	4.39	505	4.79	WITS direct
15	Meat products nec	3.14	0.14	3.14	17931	0.60	Ossa fallback
16	Vegetable oils and fats	2.00	2.00	4.98	27469	7.42	WITS direct
17	Food products nec	1.67	1.67	2.78	41729	18.13	WITS direct
18	Beverages and tobacco products	2.93	0.64	2.93	38095	7.61	Ossa fallback
19	Textiles	2.76	2.76	2.90	41684	10.75	WITS direct
20	Wearing apparel	5.31	-0.59	5.31	44040	-3.61	Ossa fallback
21	Leather products	10.21	10.21	4.11	41593	40.64	WITS direct
22	Wood products	7.42	7.42	2.29	39608	25.26	WITS direct
23	Paper products, publishing	2.73	0.29	2.73	38948	0.97	Ossa fallback
24	Chemical,rubber,plastic prods	6.45	6.45	2.37	46398	31.07	WITS direct
25	Mineral products nec	13.14	13.14	2.47	44260	49.04	WITS direct
26	Ferrous metals	8.33	8.33	3.01	43544	29.87	WITS direct
27	Metals nec	10.75	10.75	4.38	41701	30.78	WITS direct
28	Metal products	1.27	1.27	2.79	41589	5.20	WITS direct
29	Motor vehicles and parts	3.13	-1.03	3.13	40024	-7.34	Ossa fallback
30	Transport equipment nec	3.12	3.12	2.99	32600	7.64	WITS direct
31	Electronic equipment	4.25	4.25	2.49	45548	15.50	WITS direct
32	Machinery and equipment nec	2.37	0.86	2.37	45768	2.95	Ossa fallback
33	Manufactures nec	6.56	6.56	3.52	45043	31.12	WITS direct

Note: The WITS used column is the single 33-sector vector used for the main calibration. It pools valid country-triple observations from 2004, 2007, 2011, 2014, and 2017. WITS direct is the unrestricted sector estimate from the pooled WITS/MS33 regression. Ossa benchmark is the current benchmark elasticity vector used when the direct estimate fails the guardrail. Source indicates whether the row enters from the WITS direct estimate or from the Ossa fallback.

Table 30: Year-specific WITS elasticity vectors for robustness

ID	Sector	Main WITS	WITS 2004	WITS 2007	WITS 2011	WITS 2014	WITS 2017	Ossa benchmark
1	Raw and processed rice	4.87	4.87	4.87	4.87	12.19	4.87	4.87
2	Wheat	12.37	12.37	12.37	12.37	12.37	12.37	12.37
3	Cereal grains nec	3.29	3.29	3.29	3.29	3.29	3.29	3.29
4	Vegetables, fruit, nuts	2.19	2.19	2.19	2.19	2.19	1.01	2.19
5	Oil seeds	2.89	2.89	2.89	1.30	2.89	2.89	2.89
6	Raw and processed sugar	1.55	2.62	2.52	2.48	1.64	2.52	2.52
7	Plant-based fibers	10.17	4.77	4.76	16.24	17.58	23.65	2.33
8	Crops nec	2.54	2.54	2.54	2.54	1.22	1.62	2.54
9	Cattle,sheep,goats,horses	10.43	35.13	2.58	14.95	26.14	9.75	2.58
10	Animal products nec	2.12	2.12	1.32	2.12	3.49	2.12	2.12
11	Raw and processed dairy	5.60	5.60	5.60	5.60	5.60	5.60	5.60
12	Wool, silk-worm cocoons	2.40	3.46	2.47	2.89	5.27	2.89	2.89
13	Forestry	7.94	2.33	9.52	2.33	13.29	10.63	2.33
14	Meat: cattle,sheep,goats,horse	7.88	4.39	4.39	4.39	4.67	9.30	4.39
15	Meat products nec	3.14	1.80	1.23	3.14	3.14	3.14	3.14
16	Vegetable oils and fats	2.00	2.35	4.98	3.75	4.98	2.58	4.98
17	Food products nec	1.67	1.84	2.14	1.47	1.36	1.66	2.78
18	Beverages and tobacco products	2.93	2.93	2.45	2.93	2.93	2.93	2.93
19	Textiles	2.76	1.87	2.90	1.12	3.10	8.01	2.90
20	Wearing apparel	5.31	5.31	5.31	5.31	5.31	5.31	5.31
21	Leather products	10.21	6.98	14.28	12.29	8.58	9.90	4.11
22	Wood products	7.42	8.20	6.60	9.59	7.79	5.95	2.29
23	Paper products, publishing	2.73	3.86	2.73	2.73	2.73	2.27	2.73
24	Chemical,rubber,plastic prods	6.45	9.30	8.63	8.86	2.37	1.83	2.37
25	Mineral products nec	13.14	10.56	14.39	13.72	15.21	12.89	2.47
26	Ferrous metals	8.33	5.57	8.12	8.34	12.32	9.47	3.01
27	Metals nec	10.75	11.62	11.90	7.58	10.12	13.08	4.38
28	Metal products	1.27	2.15	1.39	1.31	1.09	2.79	2.79
29	Motor vehicles and parts	3.13	3.13	3.13	3.13	3.13	3.13	3.13
30	Transport equipment nec	3.12	4.57	2.63	9.29	2.99	2.99	2.99
31	Electronic equipment	4.25	7.64	4.37	4.84	2.49	2.25	2.49
32	Machinery and equipment nec	2.37	2.73	3.64	2.37	2.37	2.37	2.37
33	Manufactures nec	6.56	5.18	7.00	8.14	7.46	5.32	3.52

Note: Each year column reports the guarded WITS/MS33 sector vector estimated separately for that benchmark year. These columns are robustness diagnostics. The paper's baseline WITS calibration uses the single pooled vector in the Main WITS column.

Table 31: Selected reciprocal-result comparison across elasticity calibrations

Calibration	Scenario	Country	Bloc	Own member	Bloc-own
WITS all-year main calibration	China 2017	Argentina	0.094	0.102	-0.007
WITS all-year main calibration	China 2017	Brazil	0.131	0.129	0.002
WITS all-year main calibration	China 2017	Paraguay	0.001	0.001	0.000
WITS all-year main calibration	China 2017	Uruguay	0.598	-0.232	0.830
WITS all-year main calibration	United States 2004	Argentina	-0.077	0.062	-0.138
WITS all-year main calibration	United States 2004	Brazil	-0.111	0.047	-0.158
WITS all-year main calibration	United States 2004	Paraguay	0.490	0.200	0.290
WITS all-year main calibration	United States 2004	Uruguay	0.867	-0.006	0.874
Ossa benchmark current SIGMA	China 2017	Argentina	0.061	0.090	-0.028
Ossa benchmark current SIGMA	China 2017	Brazil	0.098	0.093	0.005
Ossa benchmark current SIGMA	China 2017	Paraguay	-0.019	-0.001	-0.018
Ossa benchmark current SIGMA	China 2017	Uruguay	0.459	-0.323	0.782
Ossa benchmark current SIGMA	United States 2004	Argentina	-0.114	0.030	-0.144
Ossa benchmark current SIGMA	United States 2004	Brazil	-0.119	0.031	-0.150
Ossa benchmark current SIGMA	United States 2004	Paraguay	0.898	0.377	0.521
Ossa benchmark current SIGMA	United States 2004	Uruguay	0.574	-0.014	0.588
GTAP pooled all-year check	China 2017	Argentina	0.063	0.098	-0.035
GTAP pooled all-year check	China 2017	Brazil	0.116	0.105	0.011
GTAP pooled all-year check	China 2017	Paraguay	-0.001	0.003	-0.004
GTAP pooled all-year check	China 2017	Uruguay	0.542	-0.401	0.943
GTAP pooled all-year check	United States 2004	Argentina	-0.073	0.075	-0.148
GTAP pooled all-year check	United States 2004	Brazil	-0.056	0.091	-0.147
GTAP pooled all-year check	United States 2004	Paraguay	0.947	0.390	0.557
GTAP pooled all-year check	United States 2004	Uruguay	0.785	-0.002	0.788
WITS early pooled robustness	China 2017	Argentina	0.100	0.108	-0.008
WITS early pooled robustness	China 2017	Brazil	0.136	0.129	0.006
WITS early pooled robustness	China 2017	Paraguay	0.004	0.001	0.002
WITS early pooled robustness	China 2017	Uruguay	0.623	-0.230	0.853
WITS early pooled robustness	United States 2004	Argentina	-0.064	0.073	-0.137
WITS early pooled robustness	United States 2004	Brazil	-0.096	0.062	-0.158
WITS early pooled robustness	United States 2004	Paraguay	0.408	0.142	0.266
WITS early pooled robustness	United States 2004	Uruguay	0.892	-0.006	0.898

Note: Bloc and Own member are real-income percentage changes relative to each calibration's same-year baseline; Bloc-own is their difference in percentage points. Differences are computed before rounding. Schedules selected under original SIGMA are held fixed across calibrations; only the two focal partner-years are reported. Paraguay China 2017 changes signs across calibrations.

Table 32: Selected zero-tariff-result comparison across elasticity calibrations

Calibration	Scenario	Country	Bloc	Own member	Bloc-own
WITS all-year main calibration	China 2017	Argentina	-0.125	-0.105	-0.020
WITS all-year main calibration	China 2017	Brazil	-0.147	-0.126	-0.020
WITS all-year main calibration	China 2017	Paraguay	-0.102	-0.086	-0.016
WITS all-year main calibration	China 2017	Uruguay	0.303	0.337	-0.034
WITS all-year main calibration	United States 2004	Argentina	-0.070	-0.035	-0.035
WITS all-year main calibration	United States 2004	Brazil	-0.105	-0.070	-0.035
WITS all-year main calibration	United States 2004	Paraguay	0.594	0.539	0.055
WITS all-year main calibration	United States 2004	Uruguay	0.950	0.948	0.002
Ossa benchmark current SIGMA	China 2017	Argentina	-0.194	-0.180	-0.014
Ossa benchmark current SIGMA	China 2017	Brazil	-0.129	-0.113	-0.015
Ossa benchmark current SIGMA	China 2017	Paraguay	-0.145	-0.140	-0.005
Ossa benchmark current SIGMA	China 2017	Uruguay	0.136	0.162	-0.026
Ossa benchmark current SIGMA	United States 2004	Argentina	-0.110	-0.065	-0.046
Ossa benchmark current SIGMA	United States 2004	Brazil	-0.116	-0.070	-0.047
Ossa benchmark current SIGMA	United States 2004	Paraguay	1.125	1.092	0.032
Ossa benchmark current SIGMA	United States 2004	Uruguay	0.618	0.695	-0.077
GTAP pooled all-year check	China 2017	Argentina	-0.251	-0.231	-0.020
GTAP pooled all-year check	China 2017	Brazil	-0.147	-0.135	-0.012
GTAP pooled all-year check	China 2017	Paraguay	-0.095	-0.080	-0.015
GTAP pooled all-year check	China 2017	Uruguay	0.152	0.171	-0.019
GTAP pooled all-year check	United States 2004	Argentina	-0.068	-0.039	-0.029
GTAP pooled all-year check	United States 2004	Brazil	-0.051	-0.015	-0.036
GTAP pooled all-year check	United States 2004	Paraguay	1.186	1.135	0.051
GTAP pooled all-year check	United States 2004	Uruguay	0.845	0.862	-0.017
WITS early pooled robustness	China 2017	Argentina	-0.164	-0.144	-0.020
WITS early pooled robustness	China 2017	Brazil	-0.164	-0.150	-0.014
WITS early pooled robustness	China 2017	Paraguay	-0.081	-0.066	-0.015
WITS early pooled robustness	China 2017	Uruguay	0.333	0.357	-0.024
WITS early pooled robustness	United States 2004	Argentina	-0.057	-0.035	-0.022
WITS early pooled robustness	United States 2004	Brazil	-0.089	-0.066	-0.023
WITS early pooled robustness	United States 2004	Paraguay	0.507	0.469	0.038
WITS early pooled robustness	United States 2004	Uruguay	0.972	0.959	0.013

Note: Bloc and Own member are real-income percentage changes relative to each calibration's same-year baseline; Bloc-own is their difference in percentage points. Differences are computed before rounding.